

XRayVision AI

A Web-Based AI-Assisted Platform for Automated X-Ray Fracture Detection, Chest Pathology Screening and Clinical Report Generation

A Final Year Project Thesis

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Science in Software Engineering (BSSE)

Submitted by

Muhammad Ali Raza — 2022F-MUL-BSSWE-017

Hamza Afzal — 2022F-MUL-BSSWE-027

Supervised by

Maam Misbah

Lecturer, School of Software Engineering

School of Software Engineering

Minhaj University Lahore

August 2026

Certificate of Approval

This is to certify that the Final Year Project titled "**XRayVision AI — A Web-Based AI-Assisted Platform for Automated X-Ray Fracture Detection, Chest Pathology Screening and Clinical Report Generation**" has been carried out by **Muhammad Ali Raza (2022F-MUL-BSSWE-017)** and **Hamza Afzal (2022F-MUL-BSSWE-027)** under my supervision, and is submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Software Engineering at the School of Software Engineering, Minhaj University Lahore.

The work presented in this thesis is, to the best of my knowledge, the original work of the candidates and has not been submitted elsewhere for the award of any other degree.

I recommend this project for evaluation by the Final Year Project Evaluation Committee.

Supervisor	Date
_____	_____
Maam Misbah	
Lecturer, School of Software Engineering	
Project Coordinator	Date
_____	_____
Head of Department	Date
_____	_____
School of Software Engineering	
External Examiner	Date
_____	_____

Declaration of Originality

We, **Muhammad Ali Raza** and **Hamza Afzal**, hereby declare that the work presented in this thesis titled **"XRayVision AI — A Web-Based AI-Assisted Platform for Automated X-Ray Fracture Detection, Chest Pathology Screening and Clinical Report Generation"** is our own original work, carried out under the supervision of Maam Misbah at the School of Software Engineering, Minhaj University Lahore.

We further declare that:

1. This thesis has not been submitted, in whole or in part, for the award of any other degree or qualification at this or any other institution.
2. All external sources — including research papers, datasets, pretrained models, open-source libraries and web resources — have been explicitly acknowledged and cited in the References.
3. Third-party pretrained machine-learning models used in this project are the intellectual property of their respective authors and are used under their published licences. Their contribution is documented in Chapter 5 and the References.
4. All source code written by us is our own. Where third-party code or components have been adapted, this is stated in the relevant section.
5. We understand that any act of plagiarism or misrepresentation is a serious academic offence and may result in disciplinary action under the regulations of Minhaj University Lahore.

_____	_____
Muhammad Ali Raza	Hamza Afzal
2022F-MUL-BSSWE-017	2022F-MUL-BSSWE-027
Date: _____	Date: _____

Dedication

To our parents, whose patience and sacrifice made this education possible.

To our teachers at Minhaj University Lahore, who taught us to build things that matter.

And to the radiologists and clinicians of Pakistan, who read more images in a day than most of us will see in a lifetime.

Acknowledgement

All praise is due to Allah, the Most Gracious, the Most Merciful, who granted us the health, patience and clarity of mind to complete this work.

We are deeply grateful to our supervisor, **Maam Misbah**, Lecturer at the School of Software Engineering, for her guidance throughout this project. Her insistence that we justify every design decision — rather than simply implement whatever was convenient — shaped both this system and the way we now approach engineering problems. The emphasis this thesis places on safety framing, graceful degradation and honest reporting of limitations is a direct result of her supervision.

We thank the faculty and administration of the **School of Software Engineering, Minhaj University Lahore**, for providing the academic environment, laboratory access and institutional support that made this project possible.

We acknowledge the open-source and research communities whose work this project stands on. In particular, the authors of **TorchXRayVision** for making clinically pretrained chest radiograph models freely available; **Ultralytics** for the YOLOv8 framework; the **Hugging Face** community for hosting and distributing model weights; and the maintainers of **FastAPI**, **React**, **TanStack** and **Supabase**. This project would not have been feasible for two undergraduate students without their generosity. We also acknowledge the **OpenStreetMap** contributors whose community-maintained data powers our clinic locator.

We are grateful to **OpenRouter** for providing free-tier access to large language models, without which the clinical synthesis component of this system could not have been built within our budget.

Finally, we thank our families and our classmates for their encouragement during the long stretches of debugging that no one sees in a finished demonstration.

Muhammad Ali Raza Hamza Afzal

August 2026

Abstract

Radiological interpretation is a bottleneck in healthcare systems worldwide, and acutely so in Pakistan, where the ratio of trained radiologists to patients is severely constrained. Medical students, meanwhile, have limited access to structured feedback while learning to read radiographs. This project addresses both problems with an educational decision-support tool rather than a diagnostic device.

This thesis presents **XRayVision AI**, a full-stack web platform that accepts a chest radiograph, a bone radiograph or an external wound photograph and returns a structured, confidence-tiered clinical report within seconds. The system implements a **routed multi-model ensemble**: rather than executing every model on every image, it dispatches the image to the specialist model matching the user's declared scan type. A DenseNet121 network pretrained on over 700,000 clinical radiographs screens chest images for eighteen pathology classes; a YOLOv8 detector localises suspected fractures with bounding boxes, supported by an image-level fracture screening classifier; and a Vision Transformer classifies external wounds. The structured outputs of these models are then passed to a large language model, which synthesises them into a readable clinical paragraph with an urgency rating, recommended actions and a specialist referral suggestion.

The platform is implemented as a React 19 single-page application built on TanStack Start, communicating over a JSON REST API with a Python FastAPI backend, and persisting to a Supabase PostgreSQL database with Row Level Security. Beyond image analysis it provides a bilingual (English/Urdu) health chatbot with voice input, a clinically guard-railed seven-day diet planner, a GPS-based clinic locator backed by OpenStreetMap, dashboard analytics, and PDF/JSON report export. A design constraint throughout was that the entire system must remain deployable on free-tier cloud infrastructure and must run inference on CPU only.

Two design principles distinguish this work from a straightforward model-serving application. The first is **safety by construction**: findings are grouped into three visible confidence tiers, any result below 60 % confidence triggers an explicit recommendation for radiologist review, negative detector outputs are never reported as evidence of normality, and an educational-use disclaimer is rendered from the application shell rather than left to individual pages. The second is **graceful degradation**: the failure of the language model, of object storage or of any single vision model downgrades the result rather than destroying it, so a user always receives whatever the system was able to determine.

The system was specified against IEEE 830, designed against IEEE 1016, and verified against a test suite of 93 specified cases documented under IEEE 829. Seventeen cases were executed against the live deployment during documentation; the remainder are specified and pending execution. Two defects remain open and are reported honestly in Chapter 6. On the deployed instance, end-to-end analysis completes in approximately 8.4 seconds on average across 31 stored scans.

The principal contribution of this work is not a new model architecture but a demonstration that clinically pretrained models, an LLM synthesis layer and disciplined safety framing can be integrated into a deployable, zero-cost web platform suitable for medical education. The thesis also documents, rather than

conceals, the limitations discovered during evaluation — including an observed label leakage issue in the fracture detector — as a contribution to anyone attempting similar work.

Keywords: medical image analysis, computer-aided diagnosis, fracture detection, chest radiography, deep learning, DenseNet121, YOLOv8, Vision Transformer, large language models, clinical decision support, FastAPI, React, educational technology

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List of Abbreviations

Abbreviation	Expansion
API	Application Programming Interface
BSSE	Bachelor of Science in Software Engineering
CLAHE	Contrast Limited Adaptive Histogram Equalisation
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
CPU	Central Processing Unit

Abbreviation	Expansion
DFD	Data Flow Diagram
DICOM	Digital Imaging and Communications in Medicine
ERD	Entity Relationship Diagram
FYP	Final Year Project
GPU	Graphics Processing Unit
HTTP / HTTPS	Hypertext Transfer Protocol (Secure)
ICD-10	International Classification of Diseases, 10th Revision
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
ML	Machine Learning
PACS	Picture Archiving and Communication System
REST	Representational State Transfer
RLS	Row Level Security
SDD	Software Design Document
SPA	Single Page Application
SRS	Software Requirements Specification
SSR	Server Side Rendering
UI / UX	User Interface / User Experience
UUID	Universally Unique Identifier
ViT	Vision Transformer
YOLO	You Only Look Once

Chapter 1: Introduction

1.1 Background

Medical imaging is one of the highest-volume diagnostic activities in modern healthcare. A chest radiograph is among the most frequently ordered imaging studies in the world, and plain radiography remains the first-line investigation for suspected fractures. Every one of these images requires interpretation by a trained radiologist or an appropriately experienced clinician.

The supply of such expertise does not scale with demand. This is a global problem, and a particularly sharp one in Pakistan, where the number of trained radiologists is small relative to population and is concentrated in major urban centres. In district and tehsil-level facilities, radiographs are frequently interpreted by clinicians whose primary training is not radiology, or are queued for remote reporting with a delay that can be clinically significant.

At the same time, deep learning applied to medical imaging has matured considerably. Since Litjens et al. surveyed the field in 2017 [11], convolutional architectures have been demonstrated at or near specialist-level performance on constrained diagnostic tasks — Esteva et al. on dermatological classification [12], Rajpurkar et al. on pneumonia detection from chest radiographs [1] and subsequently across fourteen pathology classes [13]. Large annotated datasets such as ChestX-ray8 [3] and CheXpert [4] made this progress possible, and libraries such as TorchXRyVision [5] have since made clinically pretrained weights freely available to anyone with an internet connection.

A gap remains between this research capability and its practical availability. Published models are distributed as research artefacts — checkpoints, notebooks and evaluation scripts. Turning one into something a medical student can actually use requires preprocessing pipelines, an inference service, authentication, storage, a user interface, and — critically — a presentation layer that communicates uncertainty honestly rather than presenting a probability vector as a verdict.

There is also a communication gap. A model that outputs eighteen floating-point probabilities is not useful to a third-year MBBS student. Recent large language models are capable of taking structured findings and expressing them as coherent clinical prose with recommended next steps — a translation layer between numerical model output and human clinical reasoning.

1.2 Problem Statement

Clinically capable medical imaging models exist and are freely available, but they remain inaccessible to the medical students and junior clinicians who would benefit most from them. The obstacles are not primarily algorithmic: they are the absence of an integrated, deployable system; the absence of a presentation layer that makes model uncertainty visible; and the cost of the infrastructure normally assumed by such systems.

Decomposed into specific problems:

P1 — Integration gap. Pretrained models are research artefacts, not usable applications. No preprocessing, serving, storage, authentication or interface layer is provided with them.

P2 — Interpretation gap. Raw multi-label probability outputs are not actionable for a learner. A list of eighteen numbers does not indicate what matters, how urgent it is, or what to do next.

P3 — Safety gap. Systems that present AI output as authoritative are dangerous in a clinical context. Confidence must be visible, low-confidence results must be flagged, and the absence of a detection must never be presented as proof of normality.

P4 — Cost gap. GPU inference, managed databases and commercial LLM APIs place such systems beyond the reach of a student project or a resource-constrained teaching institution.

P5 — Language gap. Health information tools in Pakistan are overwhelmingly English-only, which excludes a substantial proportion of the population who would benefit from them.

1.3 Motivation

Three observations motivated this project.

First, during our clinical-adjacent coursework we observed that medical students learning to read radiographs receive feedback slowly and unevenly. A tool offering an immediate, structured second opinion — explicitly framed as a study aid rather than an authority — fills a genuine pedagogical need.

Second, we found that the individual components required to build such a tool are all available at no cost: pretrained clinical models, free LLM inference tiers, free managed PostgreSQL, and free application hosting. What did not exist was an integration of them. Building that integration is a software engineering contribution well matched to a final year project in software engineering.

Third, we were conscious that the failure mode of a badly built medical AI tool is not a bad user experience — it is a user acting on a false negative. This gave the project an unusually clear design discipline: every

ambiguity should be resolved in the direction of visible uncertainty and explicit deferral to human expertise.

1.4 Objectives

The project set the following objectives.

Primary objective. To design, implement and deploy a web-based platform that accepts a medical image, routes it to an appropriate deep learning model, and returns a structured, confidence-tiered clinical report that is safe to use in an educational setting.

Secondary objectives.

1. To implement a routed multi-model ensemble covering chest pathology, fracture localisation and wound classification.
2. To integrate a large language model that synthesises structured model output into readable clinical prose with an urgency rating and recommended actions.
3. To design a results presentation that makes model uncertainty structurally visible.
4. To provide supporting features that extend the platform's educational utility: scan history with report export, dashboard analytics, a bilingual health chatbot, a diet planner and a clinic locator.
5. To implement authentication and per-user data isolation enforced at two independent layers.
6. To achieve all of the above within a strictly zero-cost, CPU-only deployment envelope.
7. To document the system to professional standards (IEEE 830, 1016 and 829) and to report verification results and defects honestly.

Table 1.1 — Project objectives and their success criteria

#	Objective	Success criterion
O1	Routed multi-model ensemble	Three scan types dispatch correctly to their models; the wrong model is never run
O2	LLM clinical synthesis	Every analysis returns an urgency rating, prose summary, actions and a specialist
O3	Visible uncertainty	Findings tiered by confidence; sub-60 % results trigger a review recommendation
O4	Supporting features	History, exports, analytics, chatbot, diet planner and clinic locator all functional
O5	Data isolation	A user cannot read or delete another user's data through any endpoint
O6	Zero-cost deployment	Entire system runs on free tiers with CPU-only inference
O7	Professional documentation	SRS, SDD, test documentation and user manual delivered; defects reported

1.5 Scope

Within scope. User account management; chest pathology screening across eighteen classes; fracture localisation with bounding boxes and image-level screening; external wound classification; LLM report synthesis; permanent scan history with PDF and JSON export; dashboard analytics; a bilingual English/Urdu chatbot with voice input; a seven-day diet planner with condition-specific guardrails; a GPS clinic locator; and deployment to public cloud infrastructure.

Outside scope. Regulatory certification of any kind. Integration with hospital PACS, HL7 or FHIR systems. Multi-tenant hospital administration, billing or appointment scheduling. Collaborative multi-reader review workflows. Training or fine-tuning of new models from user-supplied data. Clinical validation of diagnostic accuracy — the models are third-party pretrained artefacts used as supplied, and this project does not claim to have validated them for clinical use.

1.6 Limitations

We state the project's limitations at the outset rather than in the conclusion, because they constrain how the results in Chapter 7 should be read.

1. **The system is not a medical device** and is not certified for clinical use. It is an educational decision-support tool.
2. **Diagnostic accuracy was not independently validated.** We report the models' behaviour as integrated, not their clinical sensitivity or specificity, which would require a labelled evaluation set and clinical supervision beyond the scope of an undergraduate project.
3. **CPU-only inference** limits throughput and adds latency, particularly on cold starts.
4. **Free-tier hosting** means the backend sleeps when idle; the first request after inactivity is substantially slower.
5. **A defect in the fracture detector's label handling** was observed during evaluation and is documented in Chapter 6; it is not resolved in the submitted version.
6. **Test execution is incomplete.** Ninety-three cases are specified; seventeen were executed.

1.7 Thesis Organisation

Chapter 2 reviews the relevant literature in medical image analysis, object detection, vision transformers and LLM-assisted report generation, and compares existing systems to identify the gap this project addresses.

Chapter 3 presents the requirement analysis: elicitation method, functional and non-functional requirements, the use case model and the feasibility study.

Chapter 4 presents the system design: architecture, database design, data flow decomposition, class model, behavioural models and the design decisions with their rationale.

Chapter 5 documents the implementation: environment, backend, frontend, model integration, security, deployment, and the challenges encountered.

Chapter 6 presents the testing strategy, executed results, performance measurements and open defects.

Chapter 7 discusses the results, evaluates them against the Chapter 1 objectives, compares the system to related work, and addresses ethical and safety considerations.

Chapter 8 concludes the thesis and sets out future work.

Chapter 2: Literature Review

2.1 Deep Learning in Medical Image Analysis

The application of deep learning to medical imaging accelerated sharply after 2012. Litjens et al. [11] surveyed over three hundred contributions and identified classification, detection, segmentation and registration as the dominant task families, with convolutional neural networks the dominant architecture across all of them. Their survey also identified the field's persistent constraint: the scarcity of large, expertly labelled datasets, and the consequent reliance on transfer learning from natural-image corpora.

Esteva et al. [12] demonstrated in *Nature* that a CNN trained on clinical images could match board-certified dermatologists on skin lesion classification. The result was significant less for the architecture than for the demonstration that specialist-level performance on a constrained task was achievable with then-current methods and sufficient data — a finding that motivated similar efforts across radiology.

2.2 Chest Radiograph Classification

Progress in chest radiography was enabled by dataset releases. Wang et al. [3] published ChestX-ray8, comprising over 100,000 frontal chest radiographs labelled with eight (later fourteen) disease categories mined from radiology reports using natural language processing. Irvin et al. [4] followed with CheXpert, a dataset of over 200,000 studies whose labelling scheme explicitly modelled *uncertainty* — a design choice directly relevant to this project, since it acknowledges that radiological findings are frequently equivocal.

Rajpurkar et al. [1] introduced CheXNet, a 121-layer DenseNet trained on ChestX-ray14 for pneumonia detection, reporting performance exceeding that of practising radiologists on the F1 metric for that specific task. The architecture itself was DenseNet [2], whose dense connectivity pattern — each layer receiving the feature maps of all preceding layers — improves gradient flow and parameter efficiency, properties well suited to the relatively small datasets available in medical imaging. The authors later extended the work to CheXNeXt [13], comparing algorithmic performance to practising radiologists across fourteen pathologies.

Cohen et al. [5] made this line of work practically accessible with **TorchXRayVision**, a library providing a unified interface to multiple chest radiograph datasets and, critically, pretrained DenseNet121 weights trained across a union of large public datasets. This is the model our system uses for chest analysis. The decision to use it as supplied, rather than fine-tune it, is discussed in Chapter 4.

2.3 Object Detection and Fracture Localisation

Classification answers *whether*; fracture assessment requires *where*. Redmon et al. [6] introduced YOLO, reframing object detection as a single regression problem over a grid rather than a region-proposal pipeline, achieving real-time performance. Successive versions improved accuracy and training ergonomics; YOLOv8 [7] is the version used in this project, chosen for its mature tooling, CPU-viable inference and permissive availability.

Object detection is a natural fit for fracture assessment because a bounding box communicates *localisation* to the viewer — a student can see which region the model considered abnormal and assess that judgement against the image. This transparency is a pedagogical advantage over a whole-image classification score.

However, detection introduces a specific hazard that this project treats as a first-class design concern. A detector that outputs no positive box has not established that the study is normal; it has established only that it found nothing it recognised. Interpreting a negative detector output as evidence of normality is a category error with clinical consequences. Section 4.8 describes the design rule adopted in response.

2.4 Vision Transformers

Vaswani et al. [8] introduced the transformer architecture for sequence modelling, replacing recurrence with self-attention. Dosovitskiy et al. [9] adapted it to vision with the Vision Transformer, which partitions an image into fixed-size patches, embeds them as a sequence, and applies standard transformer encoding. ViT demonstrated that convolution is not a necessary inductive bias for image classification given sufficient data or suitable pretraining.

For wound classification — where lesion appearance is dominated by colour, texture and tissue composition rather than fine spatial structure — a ViT-based classifier is a reasonable choice, and a publicly available wound classifier [16] was adopted for this component.

2.5 Preprocessing for Radiographic Images

Radiographic images vary widely in exposure and contrast. Zuiderveld [10] introduced Contrast Limited Adaptive Histogram Equalisation (CLAHE), which performs histogram equalisation on local tiles while clipping the histogram to limit noise amplification. CLAHE is standard practice in radiographic preprocessing and is applied in our pipeline before inference, as described in Section 5.4.

2.6 Language Models for Clinical Report Generation

The translation of structured findings into clinical prose is a natural language generation task. Modern instruction-following language models can accept a structured description of findings and produce a coherent summary, an urgency assessment and recommended next steps.

This capability is genuinely useful and genuinely hazardous. A fluent, confident paragraph is persuasive independently of whether it is correct, and a language model has no access to the image — it can only reason over the findings it is given. Our system therefore constrains the model's role strictly: it receives structured findings and produces prose, an urgency label from a closed five-value vocabulary, a list of actions and a specialist suggestion. It is never permitted to introduce findings, and its output is always displayed alongside the numerical confidences it was derived from, so a reader can check the prose against the evidence.

2.7 Comparison of Existing Systems

Table 2.1 — Comparison of related systems

System / category	Task coverage	Localisation	Report synthesis	Uncertainty shown	Cost	Deployable by a student
CheXNet / CheXNeXt [1][13]	Chest only	Heatmap (CAM)	No	Score only	Research	No — research artefact
TorchXRyVision [5]	Chest only	No	No	Score only	Free	Library, not an application
Commercial radiology AI	Varies, often single-modality	Yes	Limited	Varies	High, licensed	No
General LLM chat with image upload	Broad, unspecialised	No	Yes	No	Freemium	Not purpose-built; no clinical model
Typical FYP model demonstrator	Usually one model	Sometimes	No	Rarely	Free	Local only, not deployed
XRyVision AI (this work)	Chest, fracture, wound	Yes, bounding boxes	Yes, LLM	Three visible tiers + review flag	Zero	Yes, publicly deployed

2.8 Identified Gap

Table 2.2 — Identified research and engineering gaps

Gap	Description	How this project addresses it
G1 — Integration	Capable models exist as artefacts, not applications	A complete deployed platform with preprocessing, serving, auth, storage and UI
G2 — Multi-modality routing	Existing tools handle one imaging modality	Routed ensemble across three modalities with correct model dispatch
G3 — Interpretability for learners	Probability vectors are not actionable	LLM synthesis with urgency, actions and specialist referral
G4 — Uncertainty presentation	Confidence typically shown as a bare number, if at all	Three confidence tiers, sub-60 % review flag, full confidence table
G5 — Negative-evidence safety	Detector negatives casually treated as normality	Explicit design rule: negative detector boxes are discarded, never reported as normal
G6 — Accessibility	GPU and licence costs exclude teaching institutions	Zero-cost, CPU-only architecture
G7 — Language	Health tools are English-only	Bilingual English/Urdu chatbot with RTL rendering and voice input

The contribution of this project is therefore **integrative and design-oriented rather than algorithmic**.

We do not propose a new architecture or claim improved accuracy. We demonstrate that existing clinically pretrained models, a language model synthesis layer and disciplined safety framing can be combined into a deployable system within a zero-cost envelope — and we document what breaks when you try.

Chapter 3: Requirement Analysis

3.1 Requirement Elicitation

Requirements were elicited through four activities:

1. **Domain study.** Review of the literature in Chapter 2 to establish what is technically achievable and what safety obligations follow from the domain.
2. **Supervisor consultation.** Iterative review with the project supervisor, which established the safety framing requirements and the insistence on documented design rationale.
3. **User scenario analysis.** Construction of concrete usage scenarios for each user class (medical student, clinician, allied health professional) to derive functional requirements.
4. **Constraint analysis.** Assessment of available infrastructure, which produced the hard zero-cost and CPU-only constraints that shaped the architecture.

The output was a full Software Requirements Specification conforming to IEEE 830 [17]. This chapter summarises it; the complete specification with all sixty-plus numbered requirements is provided as a companion document and in Appendix B.

3.2 User Classes

User class	Characteristics	Primary functions
Medical student (default role)	Moderate technical skill; learning radiographic interpretation	Analysis, history, chatbot
Radiologist / clinician	Domain expert, not a developer	Analysis, PDF export, clinic locator
Allied health professional	Basic technical skill	Chatbot, diet planner, clinics
Evaluator / examiner	Any	Demo account across all features
System administrator	Developer	Deployment and configuration only

The system deliberately implements **no role-based access control**. The `role` attribute is descriptive and appears in the interface and on reports, but every authenticated user has identical permissions over their own data and no permissions over anyone else's. Introducing privilege tiers without a governance model would create a false impression of clinical authorisation.

3.3 Functional Requirements

Sixty-one functional requirements were specified across nine modules.

Table 3.1 — Functional requirement groups

Module	IDs	Count	Representative requirement
Authentication and account	FR-001 – FR-010a	11	Issue HS256 JWTs expiring after 24 hours
Image analysis	FR-011 – FR-025a	16	Route the image to the model(s) matching its scan type
Results and records	FR-026 – FR-033	12	Group findings into three confidence tiers
Dashboard analytics	FR-034 – FR-037	5	Compute all statistics from the signed-in user's scans only
Health chatbot	FR-038 – FR-043a	8	Support English and Urdu with language-specific system prompts
Diet planner	FR-044 – FR-048	6	Apply condition-specific clinical guardrails
Clinic locator	FR-049 – FR-052a	6	Sort results by Haversine distance ascending
Settings	FR-053 – FR-056	4	Persist preferences server-side
Cross-cutting	FR-057 – FR-060	5	Accept requests before model loading completes

Four requirements deserve individual emphasis because they encode the project's safety position:

- **FR-021** — a chest analysis returns at most the six highest-confidence findings. Presenting all eighteen near-threshold probabilities buries the signal.
- **FR-022** — for fracture scans, only *positive* detections are reported. Negative detector boxes are discarded, because they do not establish that the study is normal.
- **FR-024** — if language model synthesis fails, the analysis still returns successfully with a fallback summary. The findings are the valuable output; losing the prose must not lose the analysis.
- **FR-028b** — if any finding falls below 60 % confidence, the results page displays an explicit recommendation for radiologist review.

3.4 Non-Functional Requirements

Table 3.2 — Non-functional requirement targets

Category	ID	Requirement	Target
Performance	NFR-P1	End-to-end analysis, warm backend	≤ 15 s
Performance	NFR-P2	Time to first accepted request after startup	≤ 5 s
Performance	NFR-P3	Non-AI endpoint latency	≤ 1 s at p95
Performance	NFR-P6	Maximum upload size	20 MB

Category	ID	Requirement	Target
Safety	NFR-S1	Educational-use disclaimer on all AI output	Always visible
Safety	NFR-S4	Negative detections never reported as normality	Absolute
Safety	NFR-S7	Sub-60 % confidence triggers review recommendation	Absolute
Security	NFR-SEC2	JWT signing	HS256, $\geq$ 32-character secret
Security	NFR-SEC3	Row Level Security	Enabled on all four tables
Security	NFR-SEC5	Rate limiting	200/min global; 10/min analyse; 20/min chat
Reliability	NFR-Q1	Graceful degradation	No single dependency failure prevents a result
Maintainability	NFR-Q5	Layer discipline	No ML logic in routers; no SQL in services
Portability	NFR-Q7	Containerised backend	Single <code>docker run</code> with an env file
Accessibility	NFR-Q11	Urgency never conveyed by colour alone	Text label always present

3.5 Use Case Model

Twelve use cases were specified with full main, alternative and exception flows.

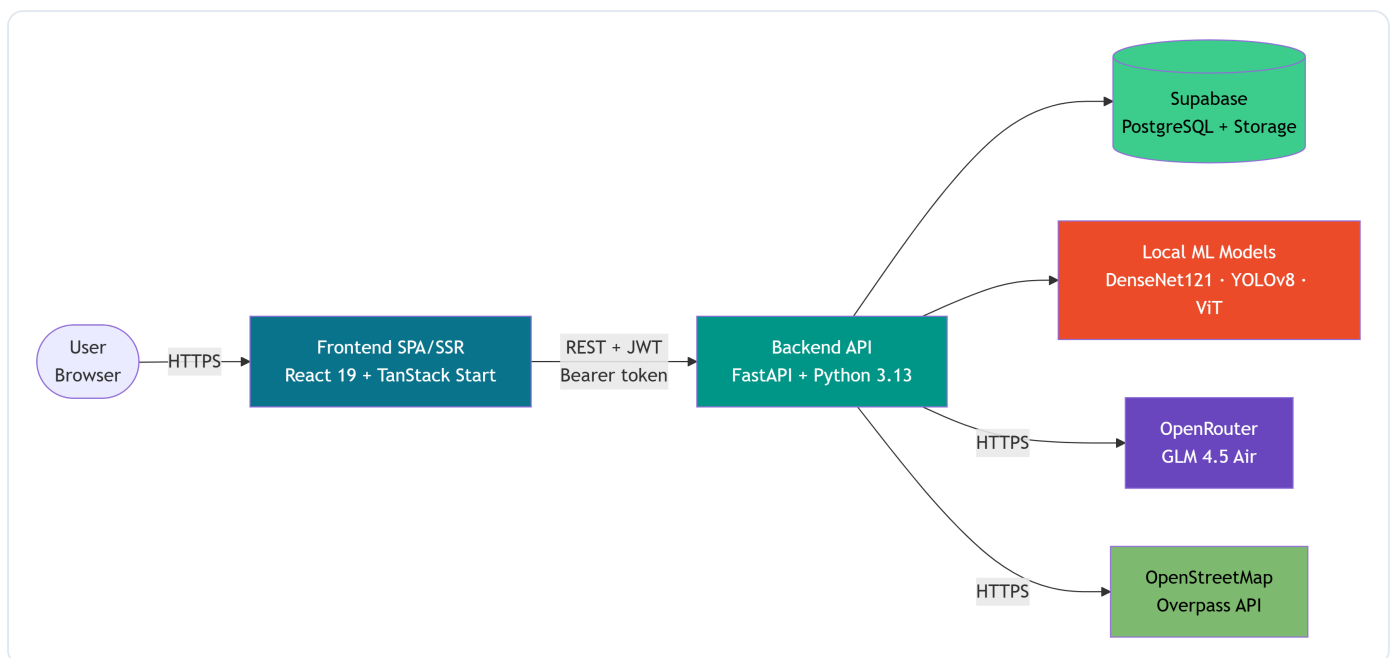


Figure 3.1 — System context: XRayVision AI and its external actors.

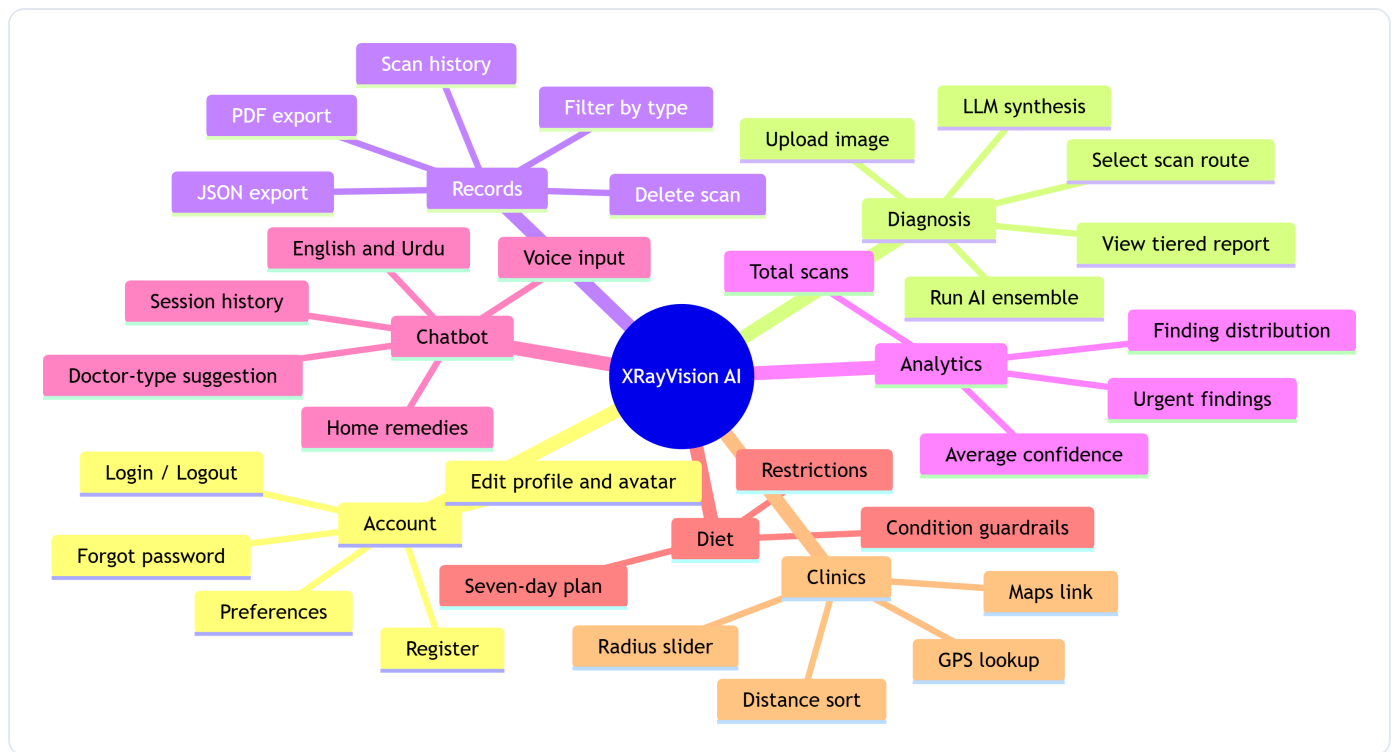


Figure 3.2 — Product function decomposition.

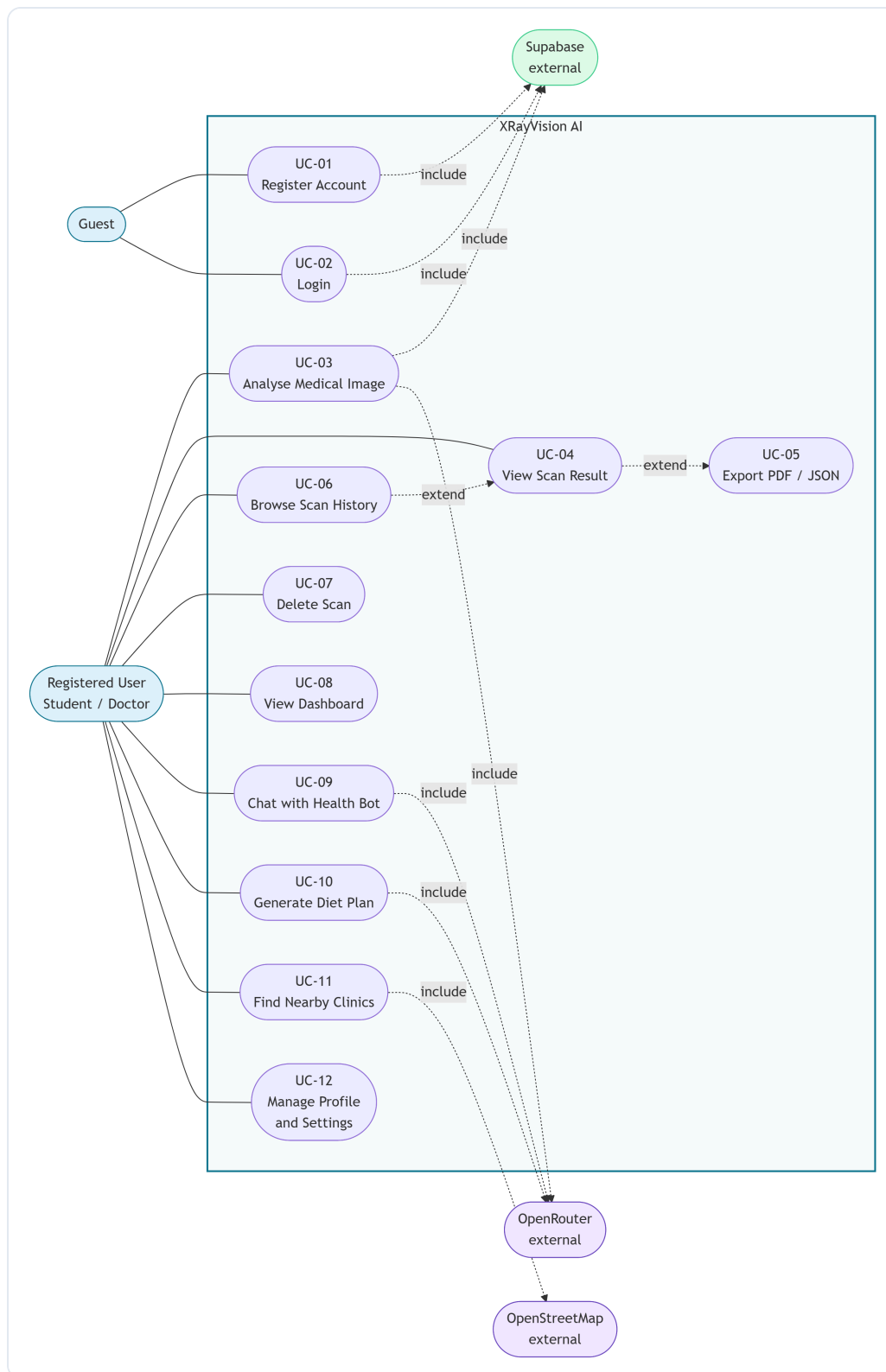


Figure 3.3 — Use case diagram, system overview.

The primary use case is **UC-03, Analyse Medical Image**. Its exception handling illustrates the system's degradation philosophy: of eight specified exception flows, only four are treated as failures. LLM synthesis failure and image upload failure are explicitly *not* errors — the flow continues and a complete result is still returned to the user.

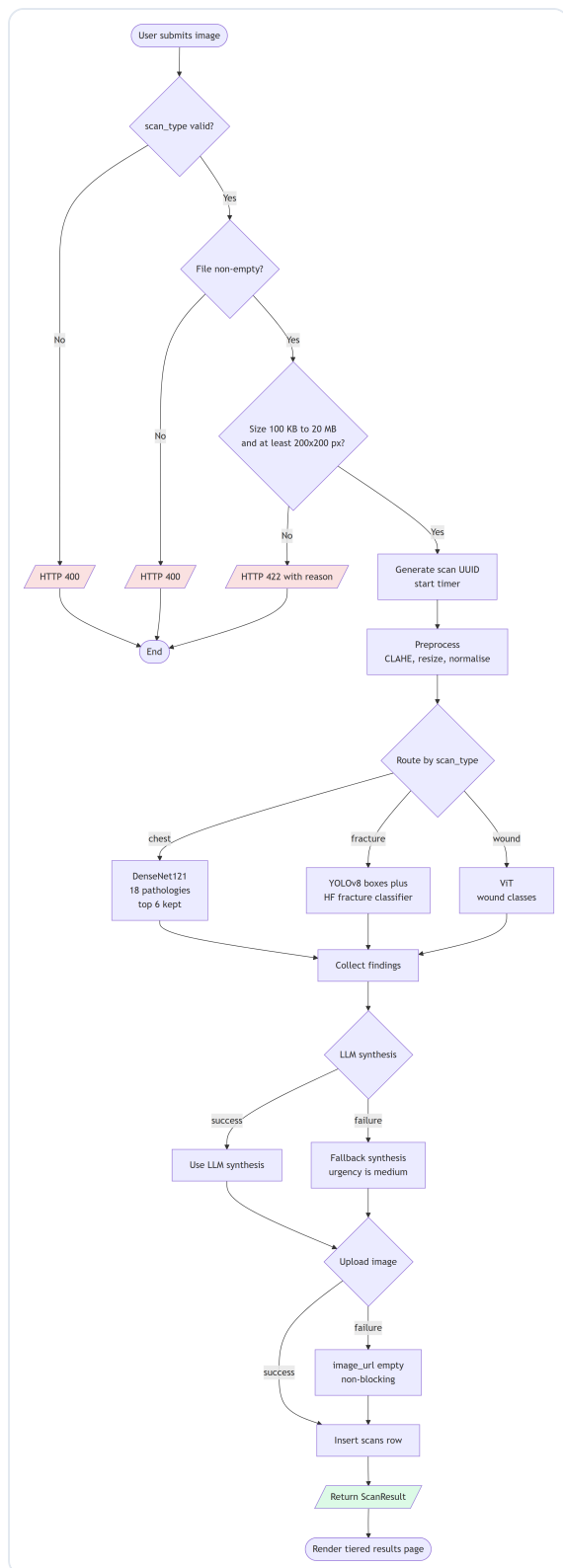


Figure 3.4 — Activity diagram for the image analysis pipeline, showing validation gates (red), degradation branches and the success path (green).

3.6 Feasibility Study

Table 3.3 — Feasibility assessment summary

Dimension	Assessment	Basis
Technical	Feasible	All required models are publicly available and run on CPU. The team had prior experience with Python, React and REST design. The principal technical risk — CPU inference latency — was mitigated by routing rather than ensembling.
Economic	Feasible at zero cost	Vercel, Hugging Face Spaces, Supabase and OpenRouter all provide sufficient free tiers. No licence, GPU or hosting expenditure was incurred.
Operational	Feasible	The target users are already fluent with web applications. The interface requires no training; the analysis workflow is three steps.
Legal / ethical	Feasible with constraints	The system must not be presented as a medical device, must carry disclaimers, and must not request or store patient-identifying information. These became requirements C-4, C-6 and NFR-S1.
Schedule	Feasible	Delivered within the two-semester FYP window. The principal schedule risk was model integration, which was front-loaded.

Chapter 4: System Design

4.1 Architectural Overview

XRayVision AI is a **three-tier layered architecture** deployed as a client–server system. Internally the backend follows a **router** → **service** → **repository** layering, with strict rules about what each layer may do.

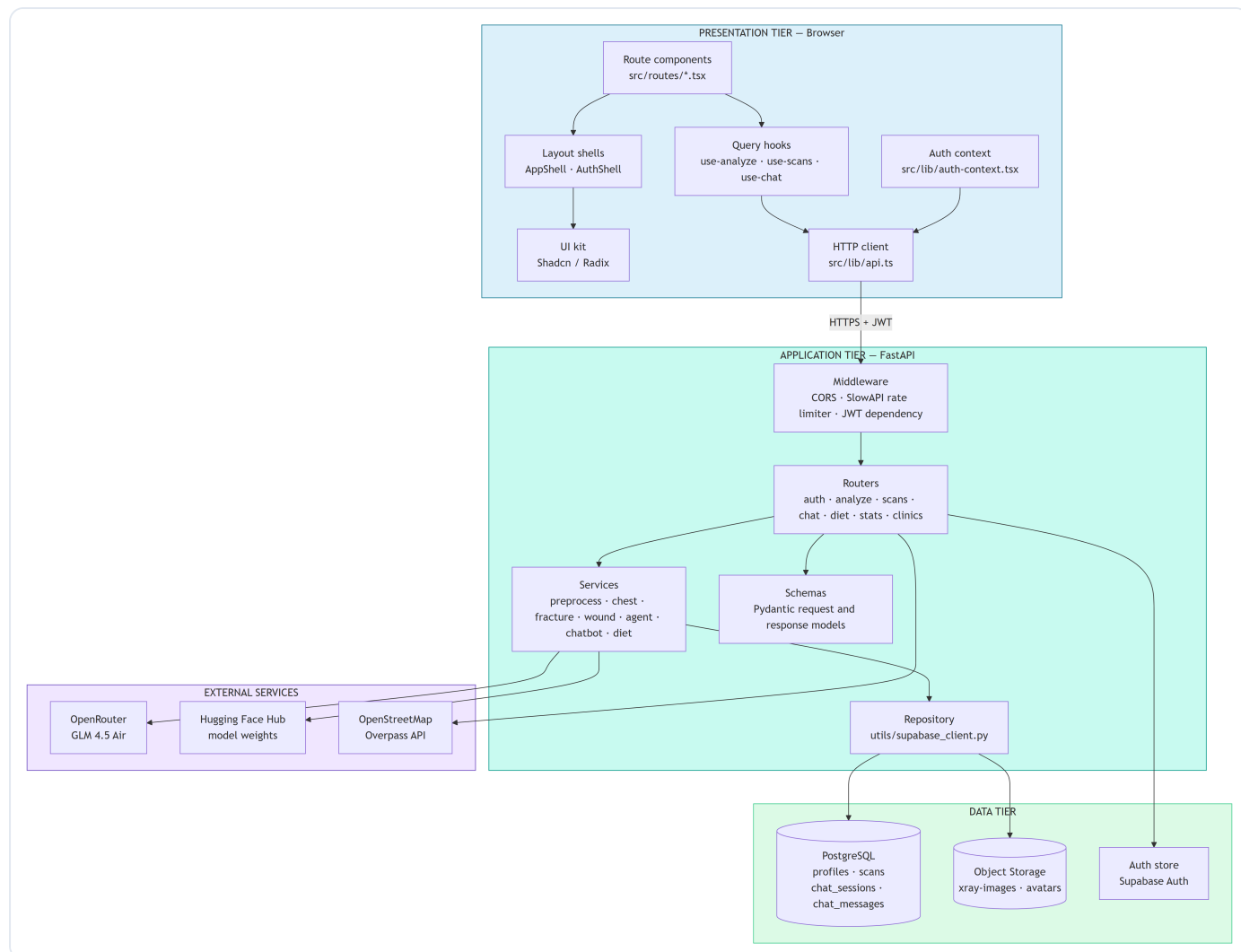


Figure 4.1 — Three-tier layered architecture with external service integrations.

Table 4.1 — Layer responsibilities and prohibitions

Layer	Module pattern	Permitted	Prohibited
Route component	<code>src/routes/*.tsx</code>	Render UI, call hooks	Calling <code>fetch</code> directly
Query hook	<code>src/hooks/use-*.ts</code>	Wrap TanStack Query around the API client	Containing business rules

Layer	Module pattern	Permitted	Prohibited
HTTP client	<code>src/lib/api.ts</code>	Build requests, attach JWT, normalise errors	Rendering anything
Router	<code>backend/app/routers/*.py</code>	Validate input, enforce auth and rate limits	Running ML, writing SQL
Service	<code>backend/app/services/*.py</code>	ML inference, LLM prompting, business rules	Knowing about HTTP
Repository	<code>backend/app/utis/supabase_client.py</code>	All database and storage access	Containing domain logic

These prohibitions are not stylistic. Keeping ML logic out of routers means every model can be invoked and tested without an HTTP client; keeping SQL out of services means the persistence technology can be replaced without touching business logic.

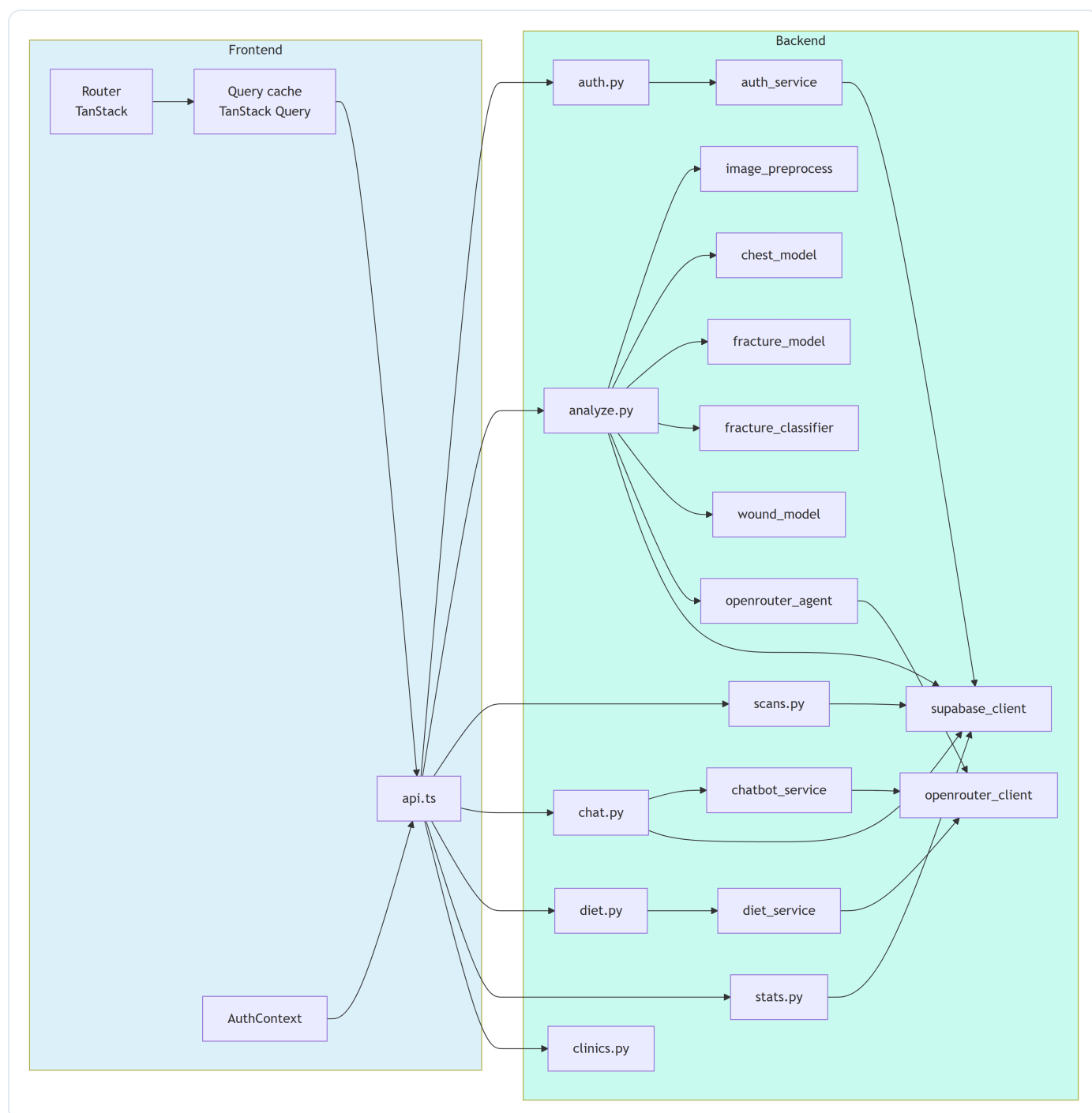


Figure 4.2 — Component diagram showing module-level dependencies.

4.2 Database Design

The data model comprises four tables. Supabase's `auth.users` table is managed by the authentication service; `profiles` extends it one-to-one.

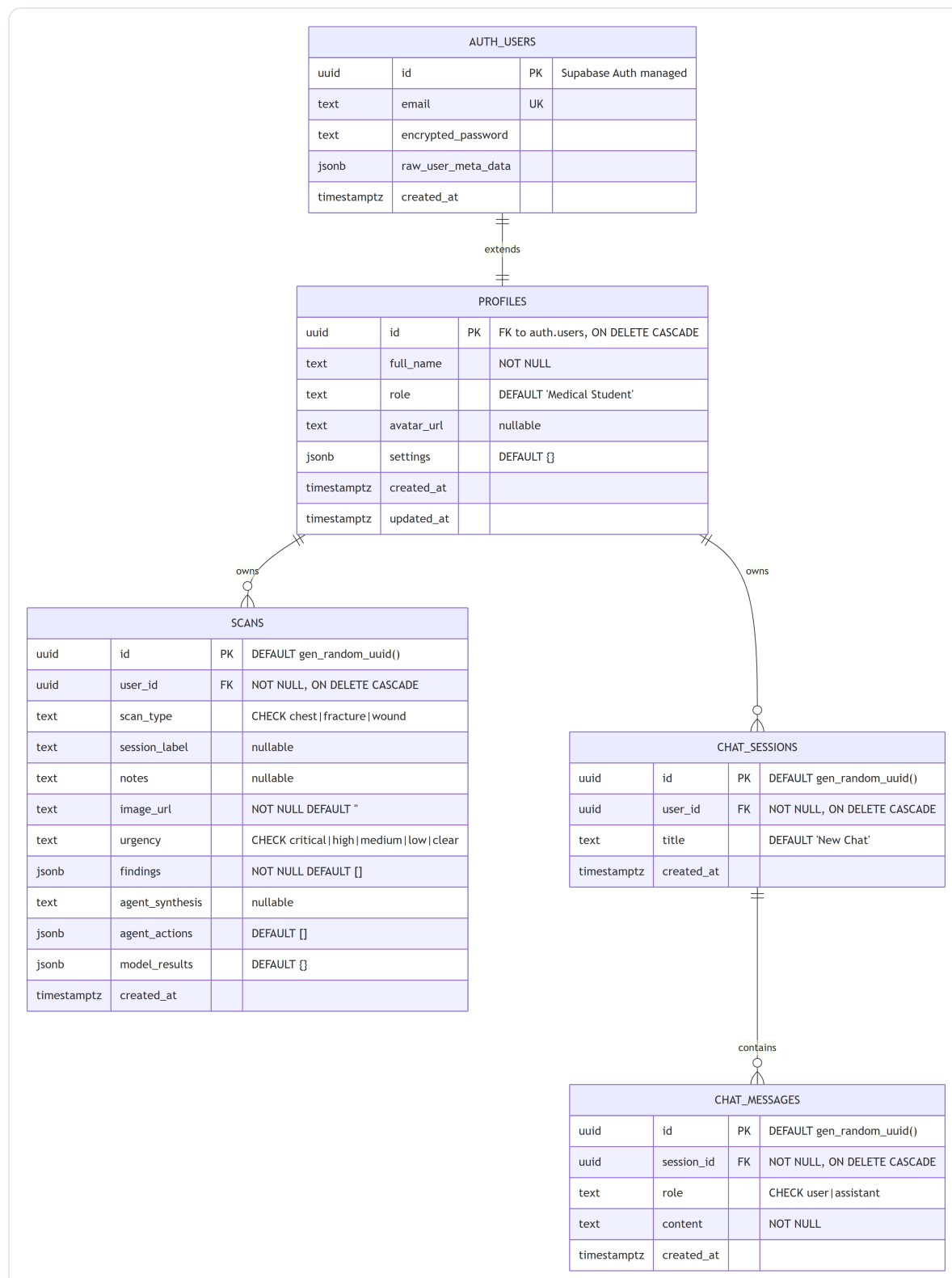


Figure 4.3 — Entity relationship diagram.

Table 4.2 — Database tables and their purpose

Table	Purpose	Key design note
profiles	Extends <code>auth.users</code> with display name, role, avatar and preferences	<code>settings</code> is JSONB so new preferences need no migration
scans	One row per completed analysis	<code>findings</code> , <code>agent_actions</code> and <code>model_results</code> are JSONB
chat_sessions	One row per chatbot conversation	Cascades from <code>profiles</code>
chat_messages	One row per message	Cascades from <code>chat_sessions</code>

Why JSONB for findings. Findings are always read as a complete set alongside their parent scan and are never queried across scans. Normalising them into a separate table would introduce a join with no functional benefit. The trade-off — losing the ability to index individual finding attributes — is accepted because no requirement needs it.

Cascade design. Every foreign key to `profiles` uses `ON DELETE CASCADE` , so deleting a user account removes their profile, scans, chat sessions and chat messages in a single operation. This satisfies the data retention requirement without application-level cleanup logic.

Four indexes support the access patterns: `user_id` on `scans` and `chat_sessions` , `created_at DESC` on `scans` for reverse-chronological history, and `session_id` on `chat_messages` .

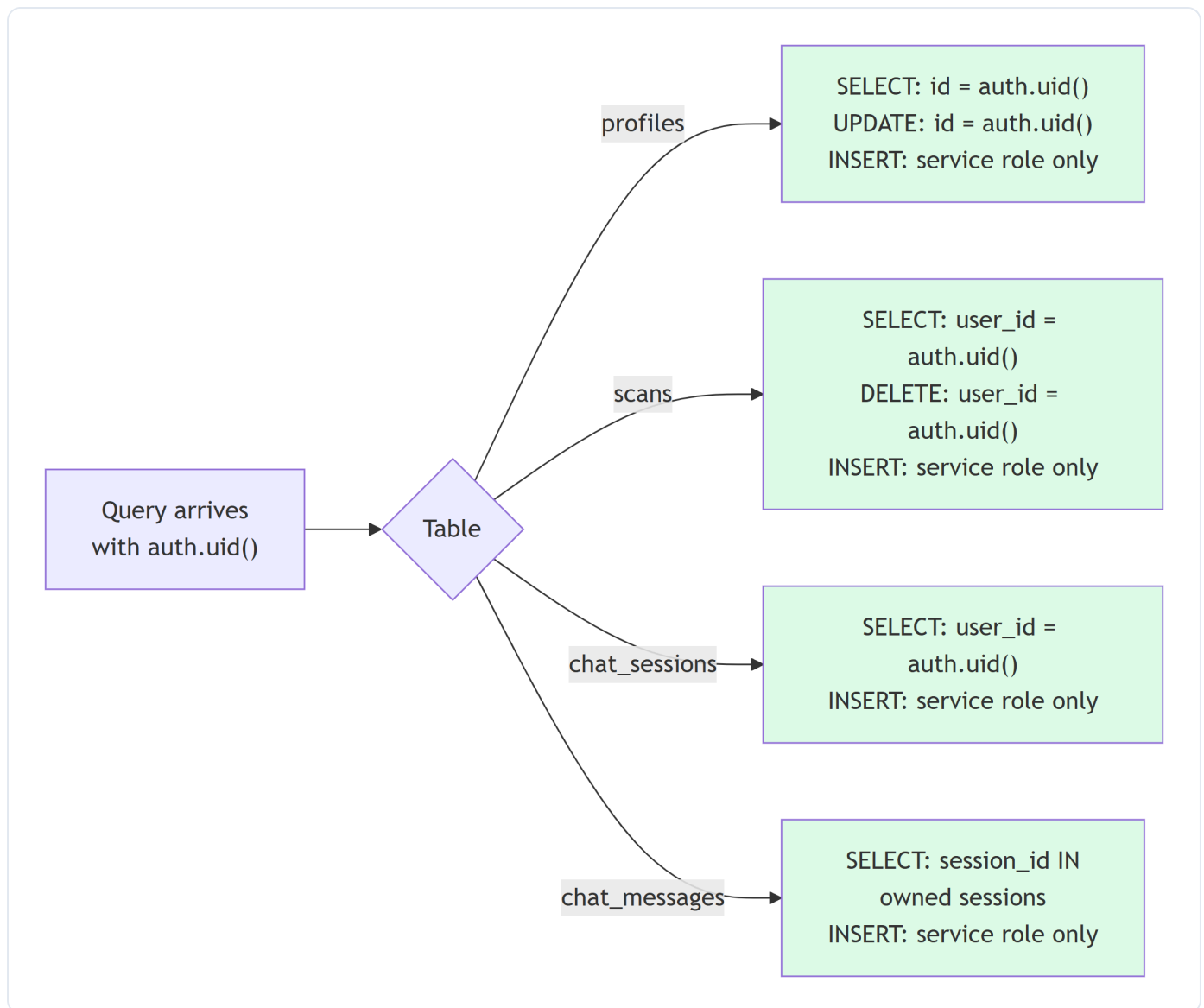


Figure 4.4 — Row Level Security policy design.

Row Level Security is enabled on all four tables. Read policies restrict `SELECT` to rows owned by `auth.uid()`. Writes are performed by the backend using the service-role key, after the backend has independently verified the JWT and resolved the owning user. Ownership is therefore checked twice — once in the application handler and once by the database — so a single missed `WHERE` clause cannot become a data breach.

4.3 Data Flow Design

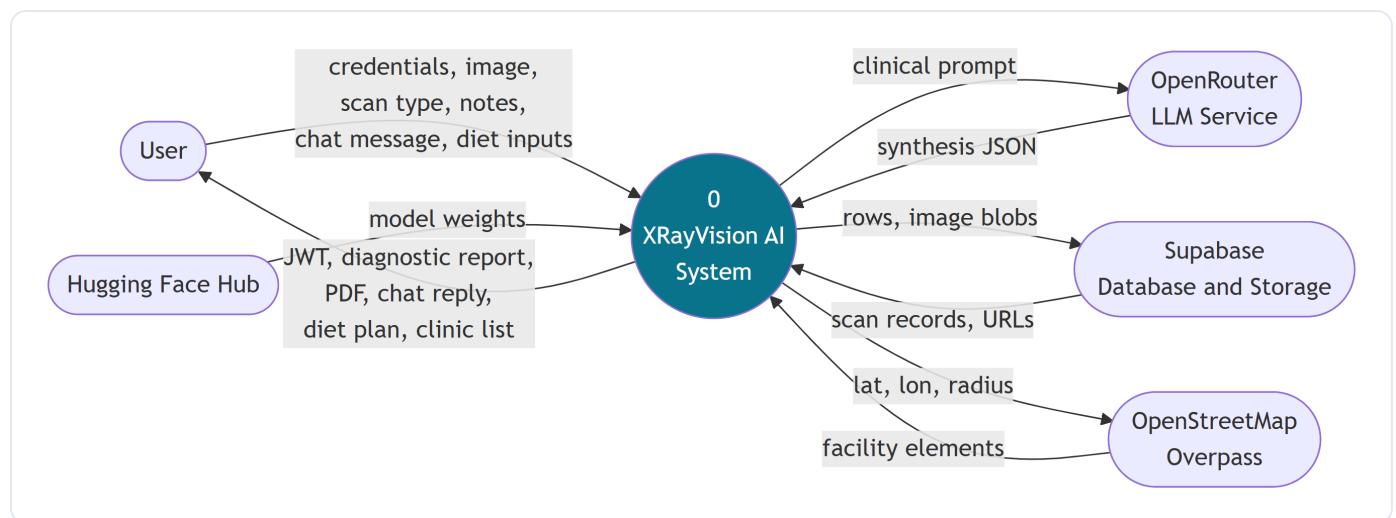


Figure 4.5 — Data Flow Diagram, Level 0 (context diagram).

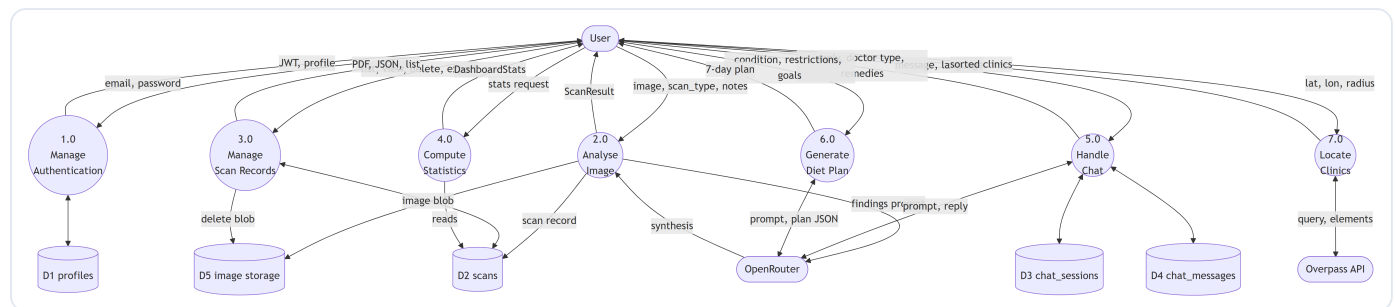


Figure 4.6 — Data Flow Diagram, Level 1: seven major processes and five data stores.

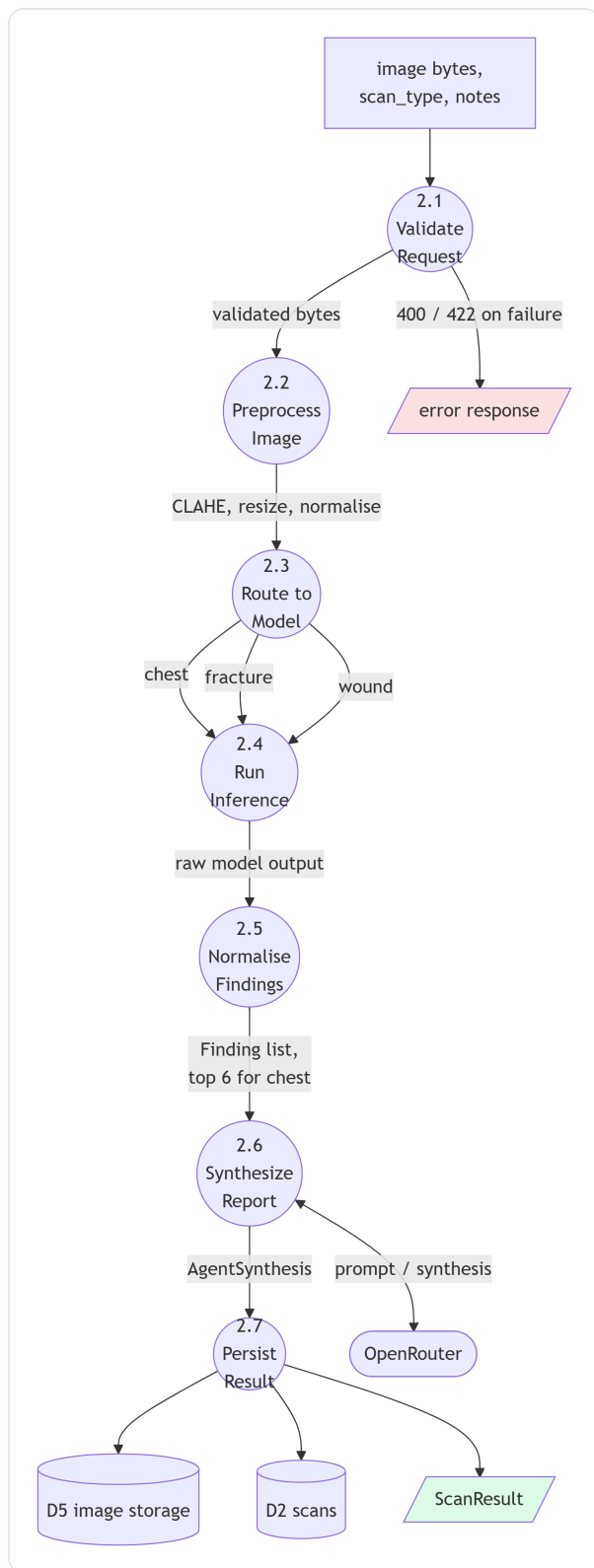


Figure 4.7 — Data Flow Diagram, Level 2: decomposition of Process 2.0, "Analyse Image", into seven sub-processes from validation through persistence.

4.4 Class Design

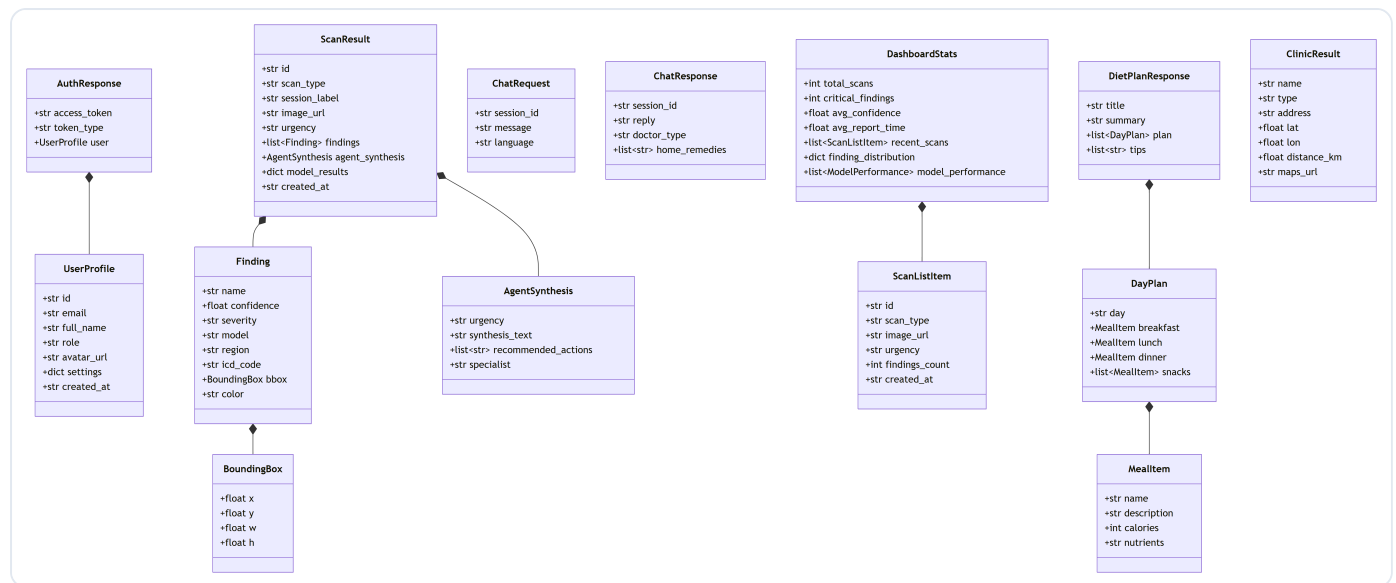


Figure 4.8 — Class diagram of the domain model as expressed in Pydantic schemas.

The domain model is defined once, in Pydantic, and mirrored in TypeScript for the frontend. **Finding** is the central abstraction: every model, regardless of architecture, must express its output as a **Finding** with a name, confidence, severity, originating model and optional bounding box. This uniform contract is what allows the routed ensemble to add or replace models without changing the results page.

4.5 Behavioural Design

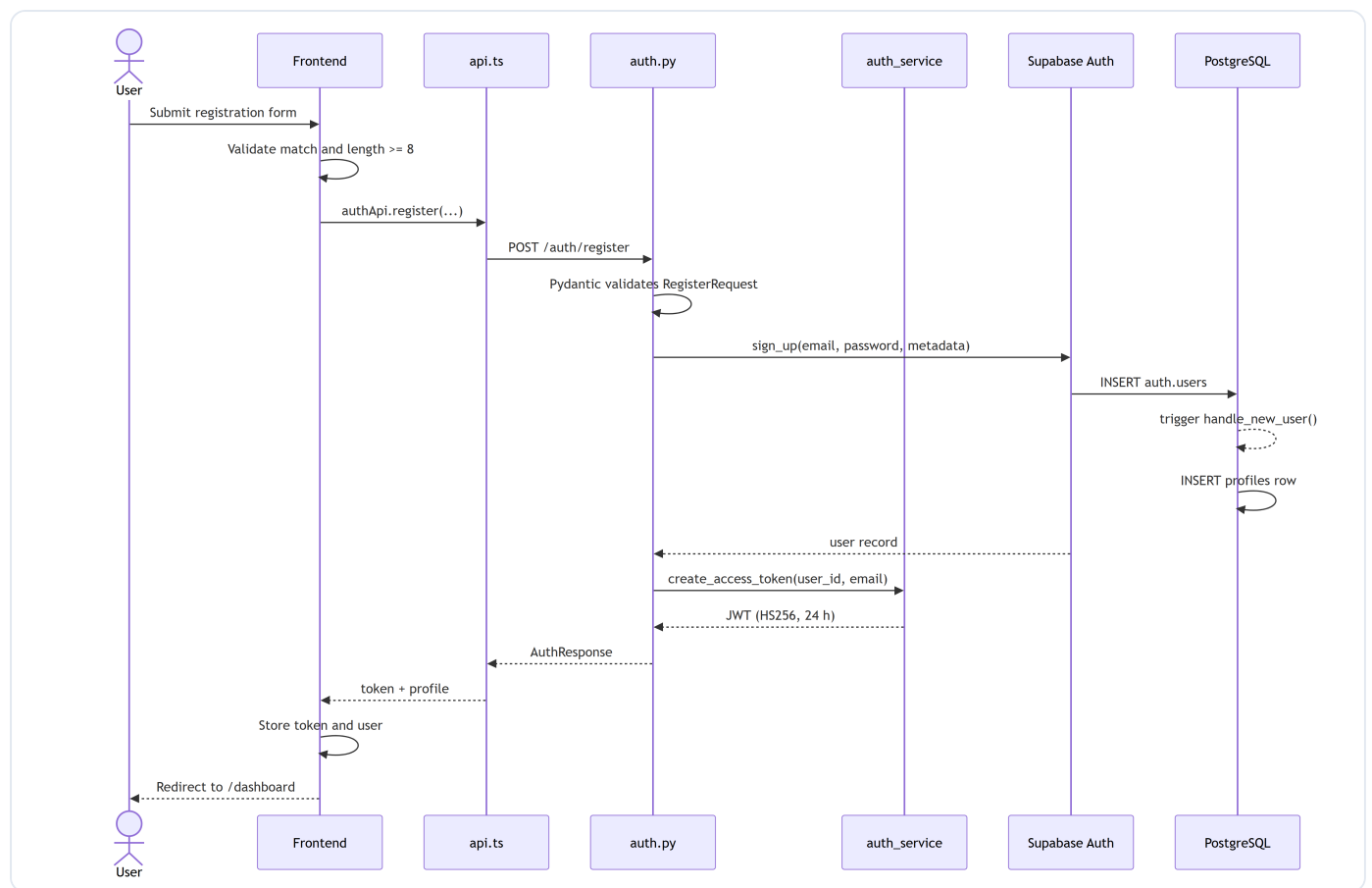


Figure 4.9 — Sequence diagram: registration and login, including the database trigger that creates the profile row.

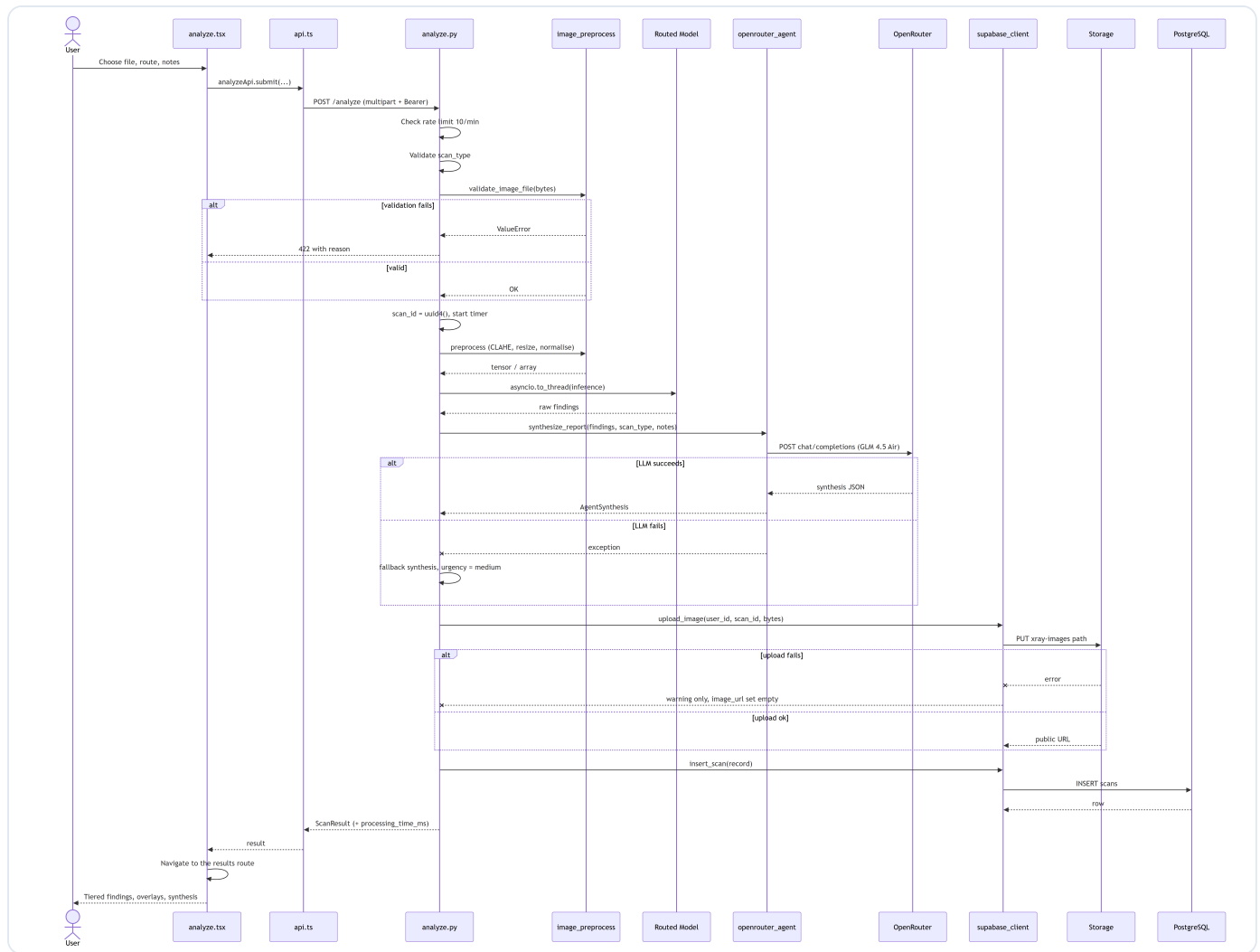


Figure 4.10 — Sequence diagram: the image analysis pipeline, showing both degradation branches (LLM failure and storage failure) that continue rather than abort.

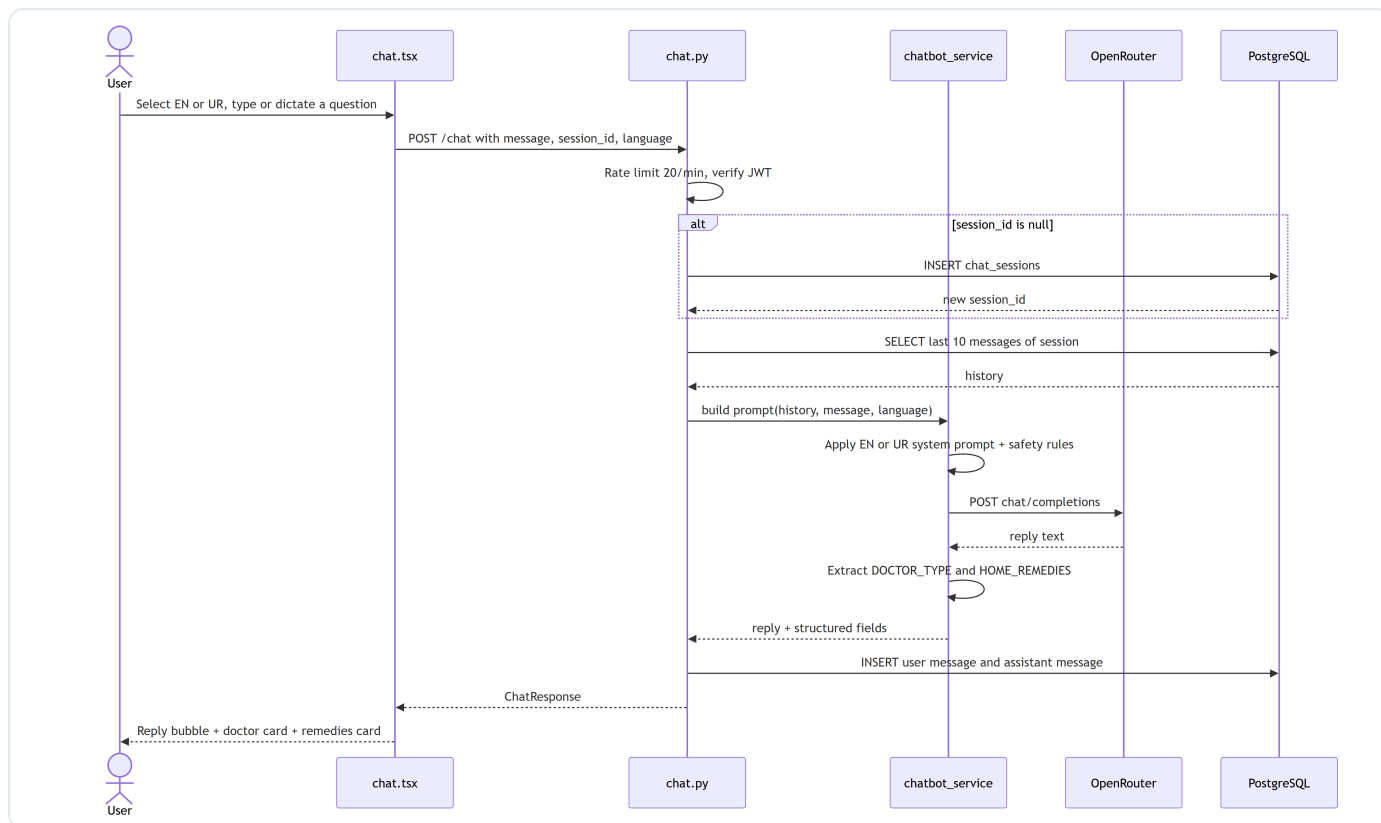


Figure 4.11 — Sequence diagram: bilingual chat with context window and structured field extraction.

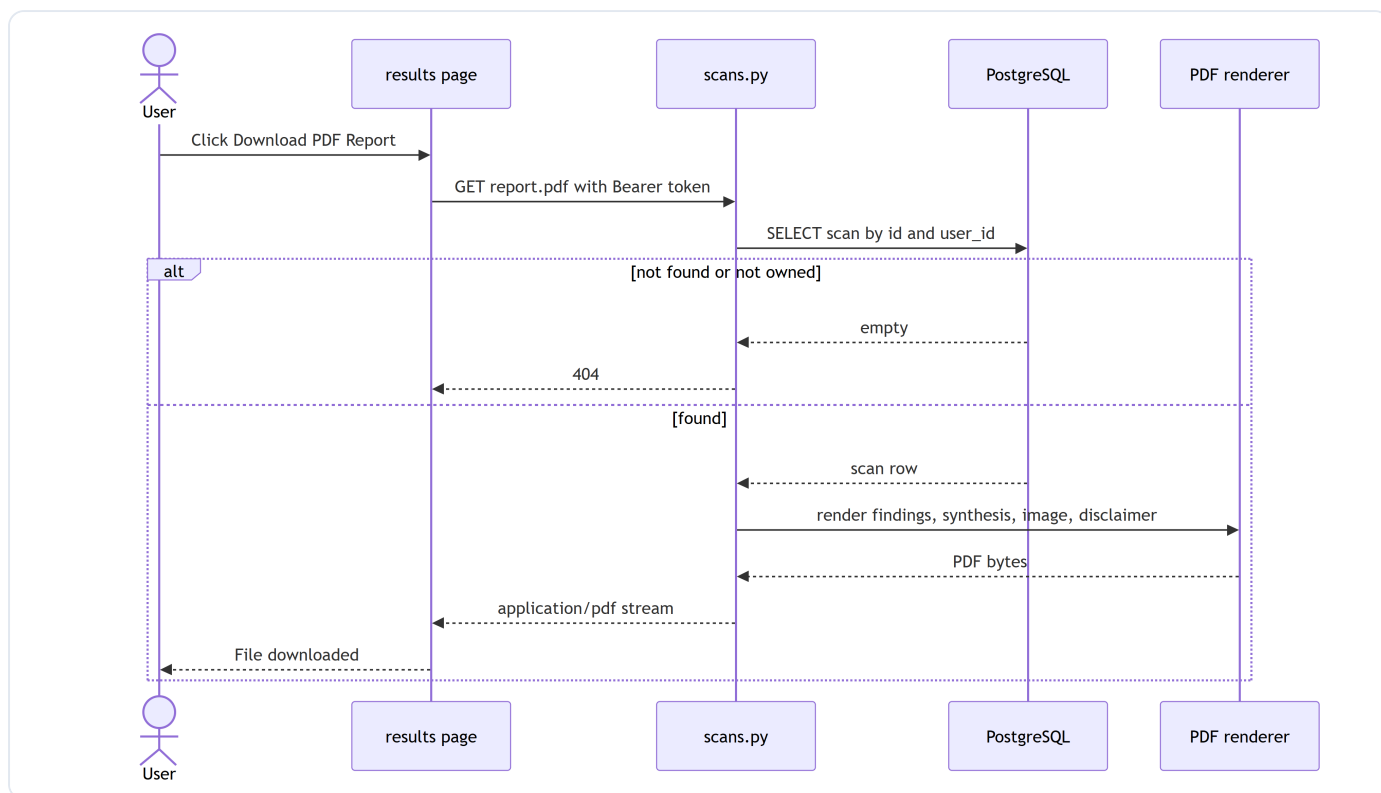


Figure 4.12 — Sequence diagram: PDF report export with ownership verification.

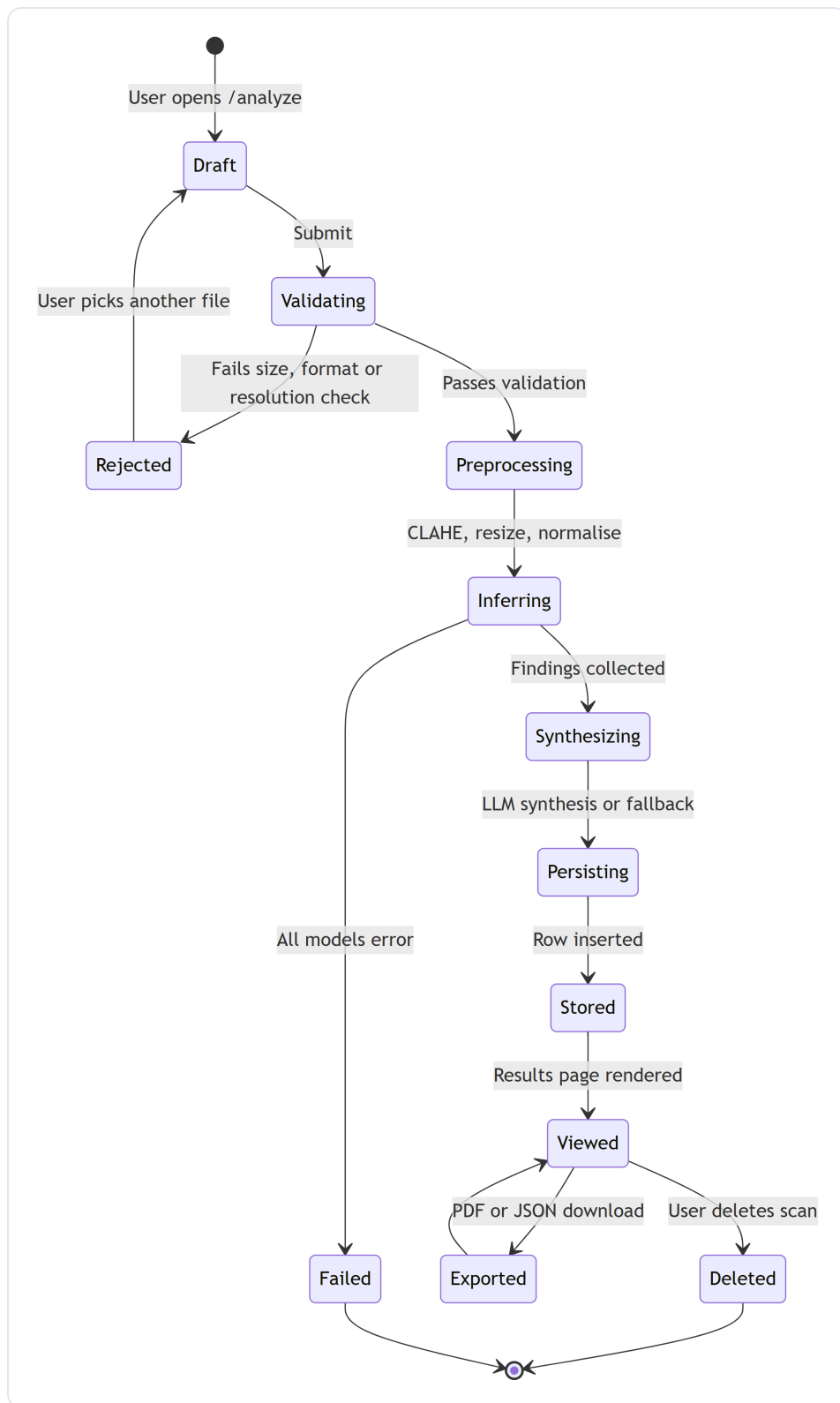


Figure 4.13 — State diagram: the lifecycle of a scan from draft to deletion.

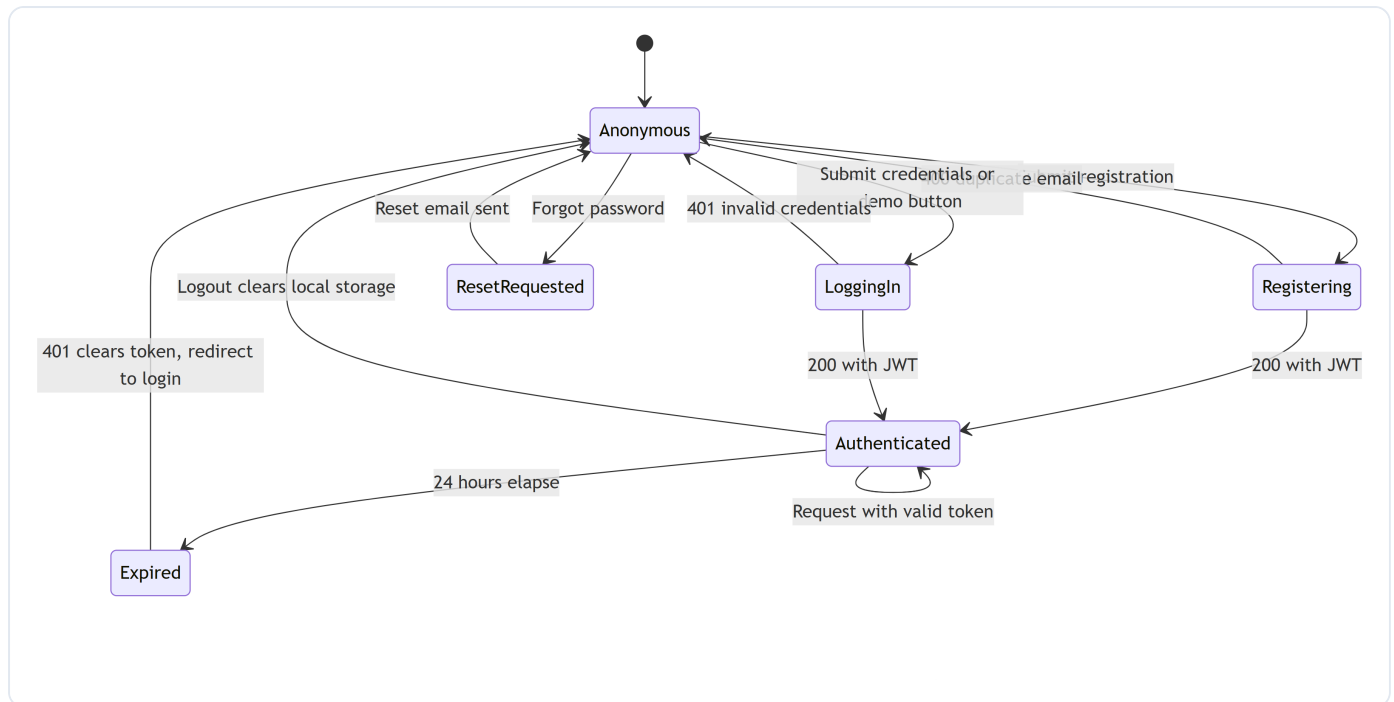


Figure 4.14 — State diagram: authentication session lifecycle including 24-hour expiry.

4.6 Interface Design

The API exposes twenty-one endpoints. All communication is JSON over HTTPS except image upload (`multipart/form-data`) and PDF export (`application/pdf`). Authenticated requests carry an `Authorization: Bearer` header.

Seven HTTP status codes carry defined meanings: 200 success; 400 malformed domain input; 401 authentication failure; 404 resource not found or not owned; 422 input failed quality validation; 429 rate limit exceeded; 500 unrecoverable server error. The distinction between 400 and 422 is deliberate — 400 signals a malformed request, 422 signals a well-formed request whose *image* failed quality checks, which is a different corrective action for the user.

4.7 User Interface Design

Two layout shells are used. `AuthShell` presents a split layout for unauthenticated pages. `AppShell` provides the authenticated layout: a collapsible sidebar with eight destinations, a header with page title and account menu, and — pinned at the bottom of the sidebar — a permanent **EDUCATIONAL USE ONLY** badge.

Placing the disclaimer in the shell rather than on individual pages is a deliberate design decision. A disclaimer that must be added to each new page will eventually be forgotten on one; a disclaimer

rendered by the shell cannot be.

The results page implements the confidence tiering that Chapter 1 identified as central: **Primary Findings** at 65 % confidence and above, **Secondary Findings** between 50 % and 65 %, and **Borderline** findings below 50 % collapsed behind an expander. A banner appears whenever any finding falls below 60 %.

4.8 Design Decisions and Rationale

Table 4.3 — Key design decisions and rationale

#	Decision	Alternative considered	Rationale
D-1	Routed ensemble	Run all models, merge outputs	A chest model applied to a wound photograph produces noise. Routing cuts CPU cost roughly threefold and keeps reports focused.
D-2	LLM synthesis layer	Display raw model scores	Multi-label probability vectors are not actionable for the target user.
D-3	Fallback synthesis on LLM failure	Return HTTP 503	The findings are the valuable output; the prose is an enhancement.
D-4	Non-blocking image upload	Abort on storage failure	The report is worth more than the thumbnail. <code>image_url</code> is nullable by design.
D-5	Discard negative detector boxes	Report them as "normal"	A negative box covers a region, not the study. Reporting it as normal would be clinically unsafe.
D-6	Background model preloading	Synchronous load at startup	Free-tier health checks time out during a multi-model synchronous load.
D-7	Stateless JWT	Server-side sessions	The backend may scale to zero; in-memory sessions would be lost on cold start.
D-8	Ownership checked in handler <i>and</i> RLS	Handler check alone	Defence in depth.
D-9	Single HTTP client module	<code>fetch</code> inside components	One place for JWT attachment, URL normalisation and 401 handling.
D-10	Confidence tiering in the UI	Flat sorted list	Uncertainty must be structurally visible, not left to the reader.
D-11	Cap chest findings at six	Report all eighteen classes	Eighteen near-threshold labels bury the signal.
D-12	Hard-coded fallback diet plan	Return malformed LLM output	Condition-specific nutrition advice must never be partially generated.
D-13	Three Overpass mirrors	A single endpoint	Public Overpass instances rate-limit aggressively.
D-14	JSONB for findings	Normalised findings table	Always read whole with the parent scan; a join would add nothing.

#	Decision	Alternative considered	Rationale
D-15	Public storage buckets	Signed URLs	Simplifies rendering while the backend may be cold. Recorded as a known limitation — see §6.5.

Chapter 5: Implementation

5.1 Development Environment and Tools

Table 5.1 — Development environment and tools

Category	Technology	Version	Role
Frontend framework	React	19	UI rendering
Frontend meta-framework	TanStack Start	—	SSR, file-based routing
Routing / data	TanStack Router, TanStack Query	—	Navigation, server state caching
Styling	TailwindCSS, Shadcn/ui, Radix	—	Design system and accessible primitives
Language (frontend)	TypeScript	5.x	Type safety
Backend framework	FastAPI	0.115	REST API, OpenAPI generation
Language (backend)	Python	3.13	Backend runtime
ASGI server	Uvicorn	—	HTTP serving
Validation	Pydantic	v2	Request/response schemas, settings
Rate limiting	SlowAPI	0.1.9	Per-IP request limiting
ML runtime	PyTorch	—	Model inference
Chest model library	TorchXRyVision	—	Pretrained DenseNet121
Detection	Ultralytics YOLOv8	—	Fracture localisation
Transformers	Hugging Face Transformers	—	ViT wound classifier
Image processing	OpenCV, Pillow, pydicom	—	CLAHE, decoding, DICOM support
Database	Supabase PostgreSQL	15+	Persistence with RLS
Storage	Supabase Storage	—	Image and avatar objects
LLM	OpenRouter GLM 4.5 Air (free tier)	—	Clinical synthesis, chatbot, diet
Frontend hosting	Vercel	—	Static + SSR deployment
Backend hosting	Hugging Face Spaces (Docker)	—	Containerised API on port 7860
Version control	Git / GitHub	—	Source management
Editor	Visual Studio Code	—	Development

5.2 Backend Implementation

The backend is organised into four packages under `backend/app/` : `routers` (HTTP boundary), `services` (business and ML logic), `models` (Pydantic schemas) and `utils` (persistence).

Application factory. `main.py` exposes `create_app()` , which configures CORS, installs the rate limiter, registers seven routers and declares two unauthenticated health endpoints. Configuration is read through a cached Pydantic `Settings` object, so no environment variable is read at more than one place in the codebase.

Startup strategy. A FastAPI lifespan handler starts model loading in a daemon background thread and returns immediately, so the server accepts requests while models are still loading.

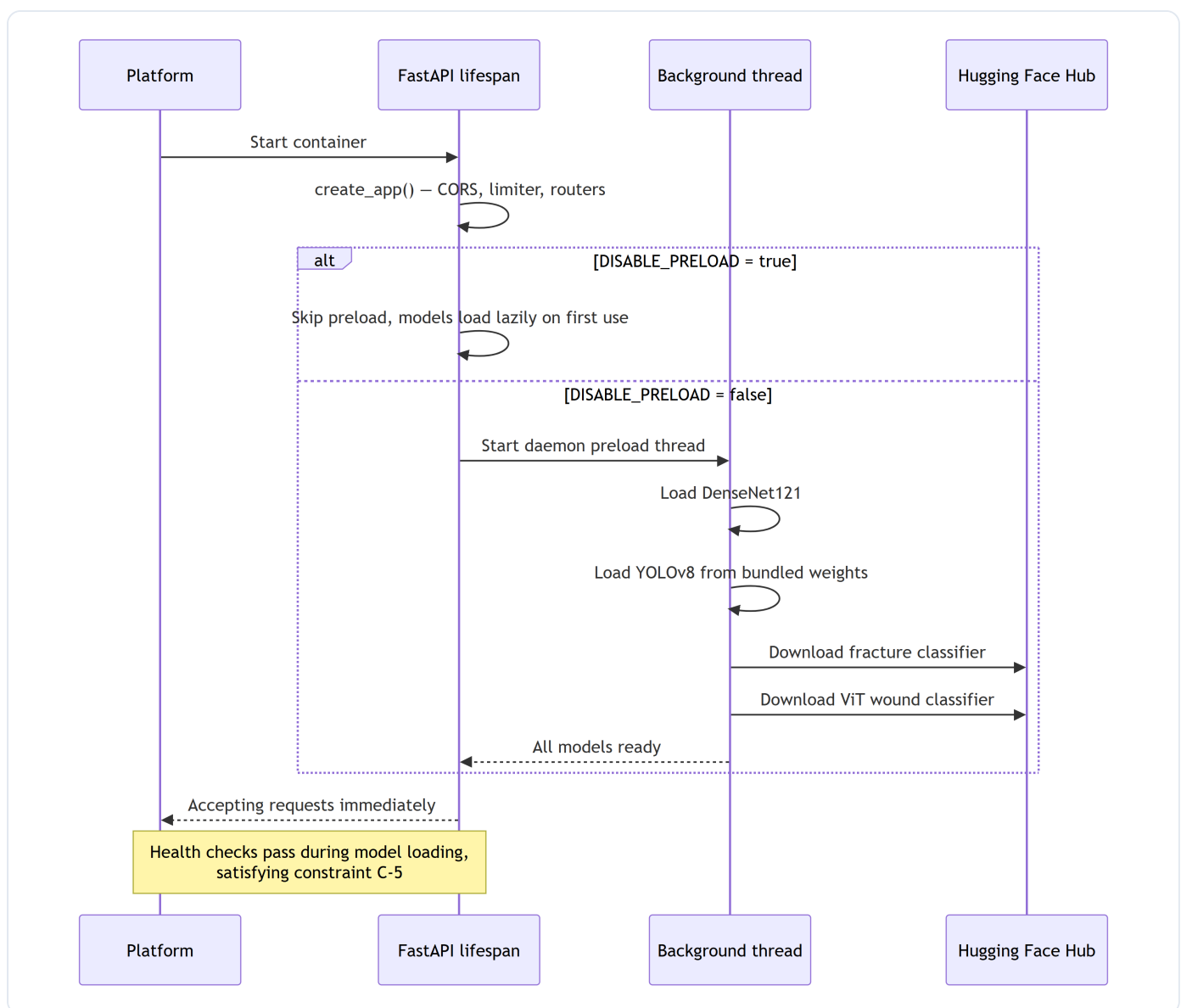


Figure 5.4 — Backend startup sequence. Health checks pass during model loading.

This was not an optimisation but a necessity: free-tier platforms apply a startup health-check timeout that a synchronous multi-model load exceeds. When `DISABLE_PRELOAD=true` , preloading is skipped entirely and

models load lazily on first use, allowing the platform to run on hosts with 512 MB of memory — authentication, chat and diet remain available even where image analysis would exhaust memory.

Rate limiting. SlowAPI applies a 200 requests/minute global default, with stricter per-endpoint limits on the LLM-backed routes.

Table 5.4 — Rate limiting configuration

Scope	Limit	Rationale
Global default	200 / minute	General abuse protection
POST /analyze	10 / minute	CPU inference is the scarcest resource
POST /chat	20 / minute	Conversational pacing, LLM quota protection
POST /diet	10 / minute	LLM quota protection

5.3 Frontend Implementation

The frontend uses file-based routing: each file in `src/routes/` maps to one URL. State is separated into two categories — server state, owned by TanStack Query and accessed through hooks in `src/hooks/`; and authentication state, held in a React context in `src/lib/auth-context.tsx`.

The single-client rule. Every network call passes through `src/lib/api.ts`. No component calls `fetch` directly. This module attaches the bearer token, normalises the base URL (stripping a trailing slash to prevent double-slash 404s), and centralises 401 handling: any 401 clears the stored token and redirects to the sign-in page, so an expired session recovers automatically.

The analysis wizard. `analyze.tsx` implements a three-step flow — upload, route selection, context — with the accepted formats and size limits stated on screen rather than discovered through an error message.

The results page. `results.$scanId.tsx` is the most involved component. It scales bounding boxes from normalised coordinates to rendered image dimensions at any viewport size, implements the three confidence tiers, provides layer toggles for findings, heatmap and labels, and supports zoom between 0.5× and 2.5×.

5.4 Preprocessing Pipeline

Every uploaded image passes through validation before any model sees it:

```
size < 100 KB          → reject (insufficient pixel data)
size > 20 MB           → reject (exceeds upload limit)
not decodable as image → reject (unless DICOM)
width or height < 200px → reject (unreliable at this resolution)
```

DICOM files bypass the PIL-based dimension check and are decoded by `pydicom` at preprocessing time. Validated images then receive CLAHE contrast enhancement [10], are resized to the target model's input dimensions and are normalised.

Rejecting low-quality input is itself a safety measure. A model will return a confident-looking score for a 150×150 pixel thumbnail; the score will be meaningless. Refusing the input is more honest than processing it.

5.5 AI Model Integration

Table 5.2 — AI model inventory

Model	Source	Task	Input	Output
DenseNet121	TorchXRayVision [5]	18 chest pathology classes	224×224 grayscale	Per-class probabilities, capped at top 6
YOLOv8	Ultralytics [7], custom weights	Fracture localisation	Original-size BGR	Positive bounding boxes only
Fracture classifier	Hugging Face [15]	Image-level fracture screening	224×224 RGB	Fracture / normal probability
ViT	Hugging Face [16]	Wound classification	RGB PIL image	Top-1 wound category
GLM 4.5 Air	OpenRouter	Clinical synthesis	Structured findings	Urgency, prose, actions, specialist

Routing. The dispatch function selects models by scan type: `chest` invokes `DenseNet121`; `fracture` invokes both `YOLOv8` and the screening classifier; `wound` invokes the `ViT`. Each invocation is wrapped individually in exception handling, and inference runs in a worker thread via `asyncio.to_thread` so the event loop is never blocked. A model that fails records its error in `model_results.model_errors` and the pipeline continues.

The two-stage fracture workflow. Fracture assessment uses detection *and* classification together. `YOLOv8` localises where it can; the classifier provides an image-level fracture suspicion independent of localisation. Critically, `Not_Fracture` detector boxes are discarded, as established in §4.8 (D-5) — a negative box covers a region, not the study.

Model models are used as supplied. No fine-tuning was performed. Fine-tuning without a properly labelled, clinically supervised dataset would risk degrading models that were trained on far more data than we could assemble, while creating a false impression of domain adaptation.

5.6 LLM Synthesis

The synthesis service constructs a prompt from the structured findings, the scan type and the user's clinical notes, and requests a response containing four fields: an urgency label from the closed set {critical, high, medium, low, clear}, a synthesis paragraph, a list of recommended actions and a specialist suggestion.

Three constraints are enforced:

1. **The model receives findings, not the image.** It cannot introduce a finding that no vision model produced.
2. **Urgency comes from a closed vocabulary.** Free-text urgency would be unparseable and incomparable across scans.
3. **Failure is non-fatal.** If the call fails or the response cannot be parsed, a fallback synthesis is substituted with `urgency: "medium"` and the recommendation to consult a radiologist.

The same client serves the chatbot and diet planner. The chatbot applies a language-specific system prompt, includes the previous ten messages as context, and extracts `DOCTOR_TYPE` and `HOME_REMEDIES` fields for display as structured cards. The diet service applies condition-specific guardrails — DASH principles for hypertension, low-glycaemic-index guidance for diabetes, and a renal-dietitian referral note for kidney disease — and validates that the returned plan contains exactly seven complete days, substituting a pre-validated fallback plan if it does not.

5.7 Clinic Locator

The locator queries the OpenStreetMap Overpass API [14] for facilities tagged `hospital`, `clinic`, `doctors`, `pharmacy` or `health_centre` within a radius, then ranks results by great-circle distance computed with the Haversine formula:

```
d = 2R * arcsin( sqrt( sin^2(dphi/2) + cos(phi1) * cos(phi2) * sin^2(dlambda/2) ) )
```

where $R = 6371$ km, ϕ = latitude in radians, λ = longitude in radians

where $R = 6371$ km. Three Overpass mirrors are attempted in sequence, because the public instances rate-limit aggressively and a single endpoint made the feature unreliable in testing.

5.8 Security Implementation

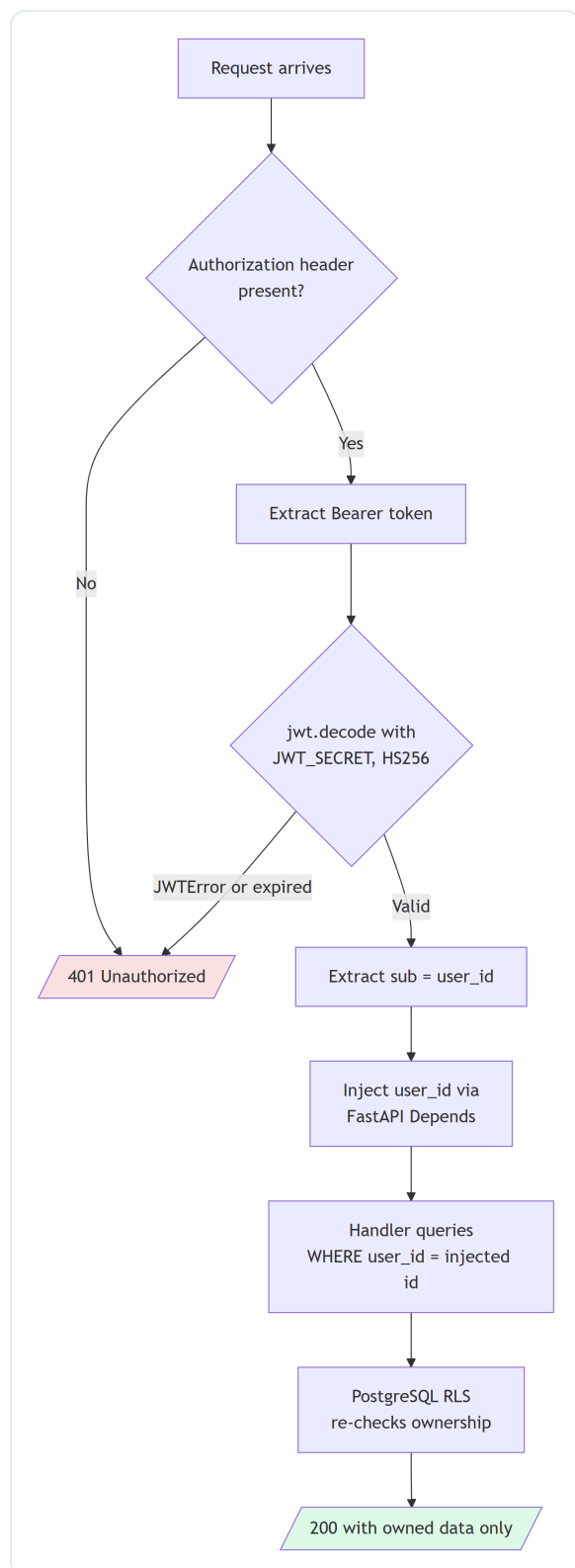


Figure 5.1 — Authentication and authorisation flow with double ownership verification.

Security is implemented in eight layers: HTTPS transport; a CORS allowlist restricted to configured origins in production; per-IP rate limiting; stateless HS256 JWTs with 24-hour expiry and a minimum 32-character secret; Pydantic validation of every request body; Row Level Security on all four tables; environment-

variable secret management; and storage paths keyed by opaque UUIDs so no patient-identifying filename is ever required.

Password hashing is delegated entirely to Supabase Auth. Implementing password storage ourselves would have added risk without adding capability.

5.9 Error Handling and Degradation

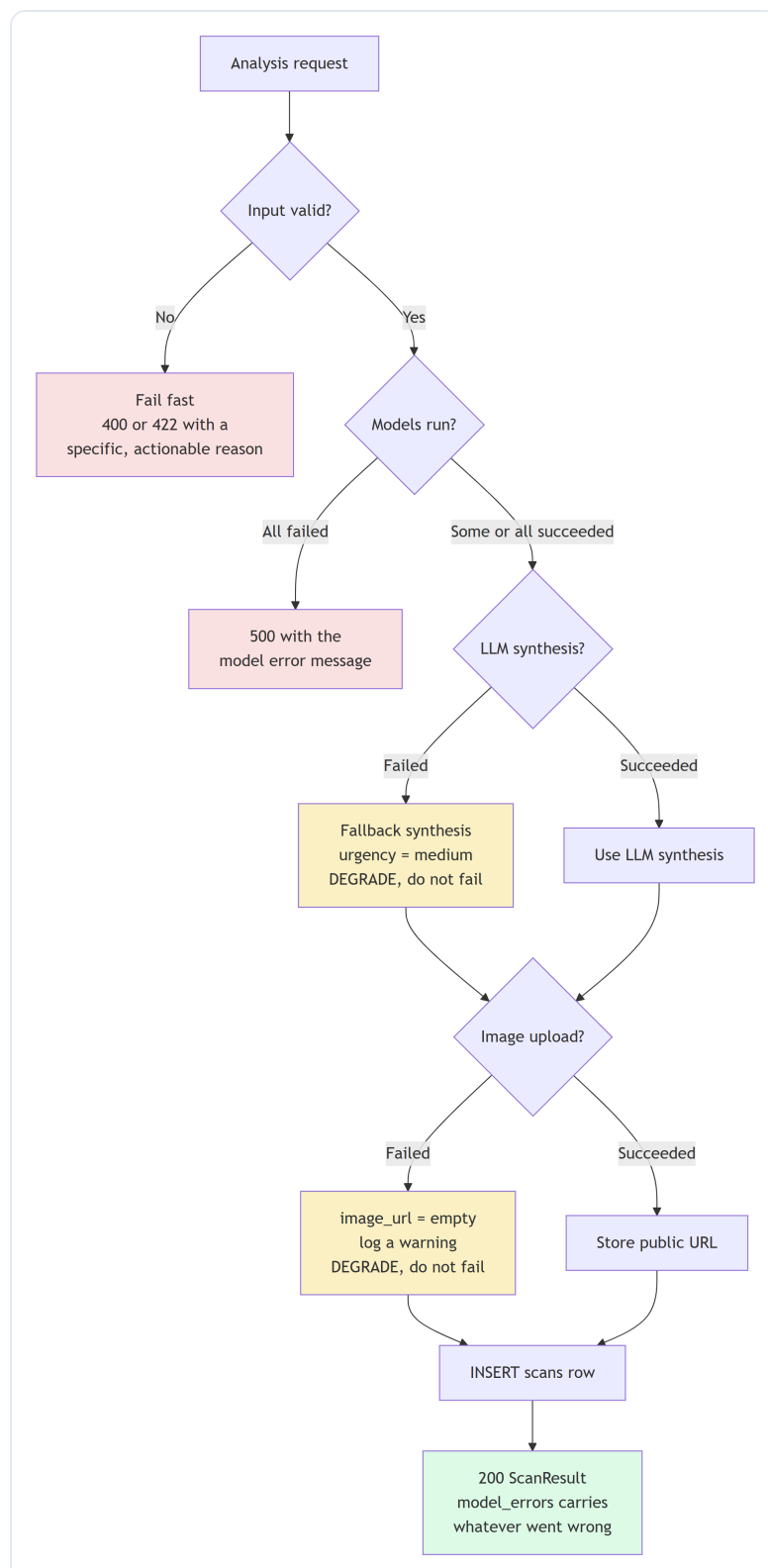


Figure 5.2 — Error handling and degradation decision tree. Red paths fail fast; amber paths degrade and continue; green is the returned result.

The governing principle is that **a partial result is more useful than an error page**. Failures are classified into three categories:

- **Fail fast** — invalid input. Return 400 or 422 immediately with a specific, actionable message.
- **Degrade** — a non-essential dependency failed. Continue and record what was lost.
- **Hard failure** — every model failed. Return 500 with the underlying error.

5.10 Deployment

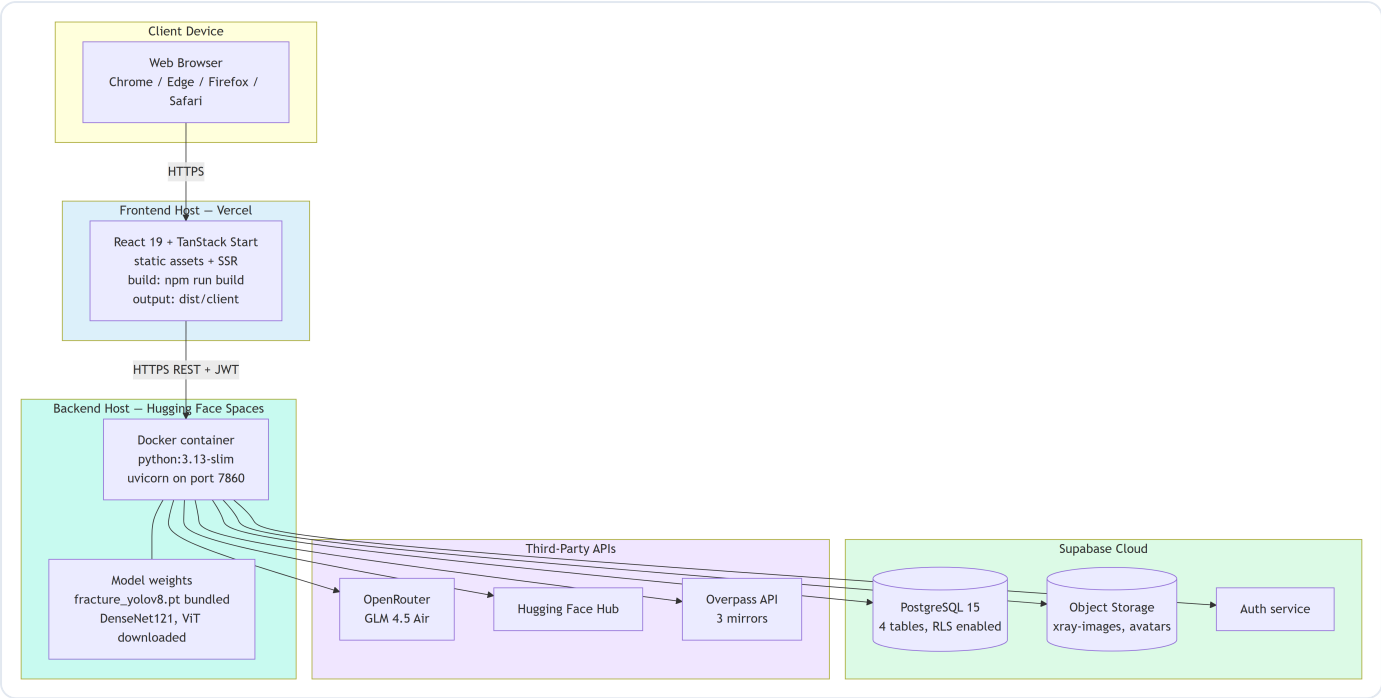


Figure 5.3 — Deployment diagram across four hosting environments.

The frontend builds to static assets plus an SSR bundle and deploys to Vercel. The backend builds to a Docker image listening on port 7860 — the Hugging Face Spaces convention — with all configuration supplied as environment secrets. The repository additionally contains a Render configuration and a Cloudflare Workers configuration, retained as portability evidence.

5.11 Implementation Challenges

Table 5.5 — Implementation challenges and solutions

#	Challenge	Solution
C1	Free-tier health checks timed out while loading four models	Background daemon thread loading; server accepts requests immediately (D-6)
C2	Memory exhaustion on 512 MB hosts	<code>DISABLE_PRELOAD</code> flag for lazy per-model loading
C3	CPU inference latency when running all models	Routed ensemble — one or two models per request instead of four (D-1)

#	Challenge	Solution
C4	Event loop blocked during synchronous inference	<code>asyncio.to_thread</code> for all model calls
C5	LLM free tier intermittently unavailable	Fallback synthesis path; analysis never fails on LLM error (D-3)
C6	Malformed LLM JSON for diet plans	Strict seven-day validation with a pre-validated fallback (D-12)
C7	Public Overpass endpoints rate-limiting	Three-mirror sequential fallback (D-13)
C8	Bounding boxes misaligned across viewport sizes	Normalised coordinates scaled at render time against actual image dimensions
C9	Double-slash 404s from a trailing slash in the API base URL	Centralised URL normalisation in the single HTTP client
C10	Avatar updates not visible after upload due to caching	Cache-busting timestamp query parameter on the returned URL
C11	Detector emitting non-anatomical class labels	Not resolved. Documented as defect DEF-03 in §6.5

Chapter 6: Testing and Evaluation

6.1 Testing Strategy

Testing was specified under IEEE 829 [19] across five levels: unit (individual service functions), integration (router → service → repository chains, exercised through the generated Swagger UI), system (end-to-end browser journeys), security (crafted requests probing authentication and isolation), and acceptance (supervisor demonstration).

Ninety-three test cases were specified, each with a requirement reference, preconditions, test data, numbered steps and an expected result. Coverage was verified against the requirements traceability matrix: **every functional requirement has at least one corresponding test case.**

Table 6.1 — Test case distribution by module

Module	Cases	Executed	Pending	Failed
Authentication	11	1	10	0
Image analysis	16	2	14	0
Scans and records	12	4	8	0
Dashboard	5	4	1	0
Chatbot	8	0	8	0
Diet planner	6	0	6	0
Clinic locator	6	2	4	0
Settings and profile	4	0	4	0
System	5	0	5	0
Security	9	0	8	1
Performance	5	0	5	0
Usability	6	4	2	0
Total	93	17	75	1

6.2 Statement on Execution Status

We report execution status without inflation. Of ninety-three specified cases, **seventeen were executed** against the live deployment using an automated browser session signed in to the demonstration account,

with screenshot evidence retained for each. **Seventy-five remain pending** — principally those requiring real medical image fixtures, deliberate negative-path inputs, instrumented timing measurements, or multiple accounts for isolation testing. **One case failed**, and is reported in §6.5.

A test case that has not been run is not a passing test case. Presenting the specification as though it were an execution report would be a misrepresentation, and we have not done so.

6.3 Executed Test Results

Table 6.2 — Executed test results

Test ID	Requirement	Verified behaviour	Result
TC-AUTH-05	FR-005, FR-006	Valid credentials authenticate and land on the dashboard with populated data	Pass
TC-ANLZ-02	FR-020, FR-022	Fracture scan renders bounding boxes; no <code>Not_Fracture</code> finding appears in the response or UI	Pass
TC-ANLZ-12	FR-023	A scan with no findings above threshold stores urgency <code>clear</code> and renders an empty-findings state	Pass
TC-SCAN-01	FR-029	History lists the user's scans in reverse-chronological order	Pass
TC-SCAN-03	FR-027, FR-028	Opening a scan renders image, overlays, findings and synthesis for that scan	Pass
TC-SCAN-04	FR-028a	Findings grouped as Primary ($\geq 65\%$), Secondary (50–65 %) and collapsed Borderline ($< 50\%$)	Pass
TC-SCAN-05	FR-028b	A sub-60 % finding triggers the radiologist-review banner	Pass
TC-DASH-01	FR-034	Four headline statistics render with correct values	Pass
TC-DASH-02	FR-035	Finding distribution renders as a ranked frequency list	Pass
TC-DASH-03	FR-036	Recent analyses list links through to full reports	Pass
TC-DASH-04	FR-036a	Per-model average confidence renders for all three vision models	Pass
TC-CLIN-02	FR-049	Radius slider spans 1–20 km with a 5 km default	Pass
TC-CLIN-04	UI-5	Pre-search empty state renders with OpenStreetMap attribution	Pass
TC-UI-01	UI-1	All six landing page sections render without blank regions	Pass

Test ID	Requirement	Verified behaviour	Result
TC-UI-03	NFR-S1, LR-1	Educational-use badge present in the shell on every authenticated page; full disclaimer on reports	Pass
TC-UI-04	NFR-Q3	Analysis wizard states accepted formats and size limits on screen	Pass
TC-UI-05	UI-4, NFR-Q11	Every urgency badge carries a text label alongside its colour	Pass
TC-SEC-09	NFR-SEC8	Repository scanned for tracked secrets	Fail — see DEF-01

The safety-critical behaviours verified here are worth emphasising: TC-ANLZ-02 confirms the negative-detection rule (D-5) holds in the deployed system; TC-SCAN-04 and TC-SCAN-05 confirm that confidence tiering and the low-confidence review flag operate as designed; and TC-UI-03 confirms the disclaimer is genuinely unconditional.

6.4 Observed System Metrics

Measurements taken from the deployed instance's dashboard across 31 stored analyses:

Table 7.1 — Observed system metrics on the deployed instance

Metric	Observed value
Total stored analyses	31
Analyses rated high or critical urgency	15
Mean finding confidence across all scans	63.2 %
Mean end-to-end report time	8.4 s
Mean DenseNet121 confidence	84.7 %
Mean YOLOv8 confidence	89.1 %
Mean ViT confidence	82.3 %

The 8.4-second mean report time satisfies target NFR-P1 (≤ 15 s) on a warm backend. Most frequent findings across the stored set were Lung Opacity (10 occurrences), Lung Lesion (8), and Nodule, Mass and Enlarged Cardiomedastinum (3 each).

These are operational metrics from the system's own analytics, not a controlled experiment. They describe the system's behaviour on the images it has processed; they are **not** measurements of diagnostic accuracy, which was outside this project's scope (§1.6).

6.5 Defects

Table 6.3 — Open defects

ID	Severity	Description	Status
DEF-01	Medium	The root <code>.env</code> file is tracked in version control. It contains the API base URL, the Supabase project URL and the Supabase publishable anon key. No service-role key, LLM key or JWT secret is exposed, so this is a hygiene defect rather than a credential compromise — but the file is listed in <code>.gitignore</code> and should not be tracked. Remediation: <code>git rm --cached .env</code> .	Open
DEF-02	Low	<code>backend/README.md</code> instructs the operator to create the <code>xray-images</code> storage bucket as <i>private</i> , while the implementation creates it with <code>public: True</code> and serves images through public URLs. The root README is correct; the backend README contradicts it. Remediation: correct the backend README.	Open
DEF-03	Medium	The fracture detector was observed emitting a non-anatomical class label (<code>Vase</code>) as a low-confidence secondary finding on a wrist radiograph, alongside a correct high-confidence fracture finding from the screening classifier. Root cause identified: the bundled <code>backend/models/fracture_yolov8.pt</code> is correct — its class vocabulary is <code>Fracture / Not_Fracture</code> and contains no COCO labels. A generic COCO checkpoint, <code>backend/yolov8n.pt</code> , was also present in the repository, and its vocabulary does contain <code>vase</code> , <code>person</code> and <code>toothbrush</code> . The deployed instance was therefore running the generic checkpoint rather than the fracture model. The <code>ALLOW_GENERIC_YOLO_WEIGHTS</code> guard did not prevent this because it matches on the weight <i>filename</i> only, so a generic checkpoint reached through a misconfigured <code>YOLO_WEIGHTS_PATH</code> still loads. Impact: non-clinical labels appear in the Secondary and Borderline tiers. Confidence tiering limited the harm — the spurious labels appeared at 52.7 % and below while the correct finding held the Primary tier — but the behaviour is incorrect. Remediation: verify <code>YOLO_WEIGHTS_PATH</code> and <code>ALLOW_GENERIC_YOLO_WEIGHTS</code> on the deployed backend; strengthen the guard to inspect the loaded model's class vocabulary instead of its filename; the stray <code>yolov8n.pt</code> has been removed from version control.	Open

DEF-03 is the most substantive finding of the evaluation, and we report it deliberately. It illustrates a general hazard in multi-model systems: a model integrated correctly at the API level can still be wrong at the *semantic* level, and no amount of exception handling detects a confidently-returned nonsense label. It also shows that a filename-based safety guard is not a safety guard: the only reliable check is to inspect what the loaded model actually predicts. It also demonstrates the value of the confidence tiering design — the spurious output was automatically relegated below the correct finding rather than presented as equally authoritative.

Chapter 7: Results and Discussion

7.1 Delivered System

The system is publicly deployed and operational. The following figures show the delivered interface.

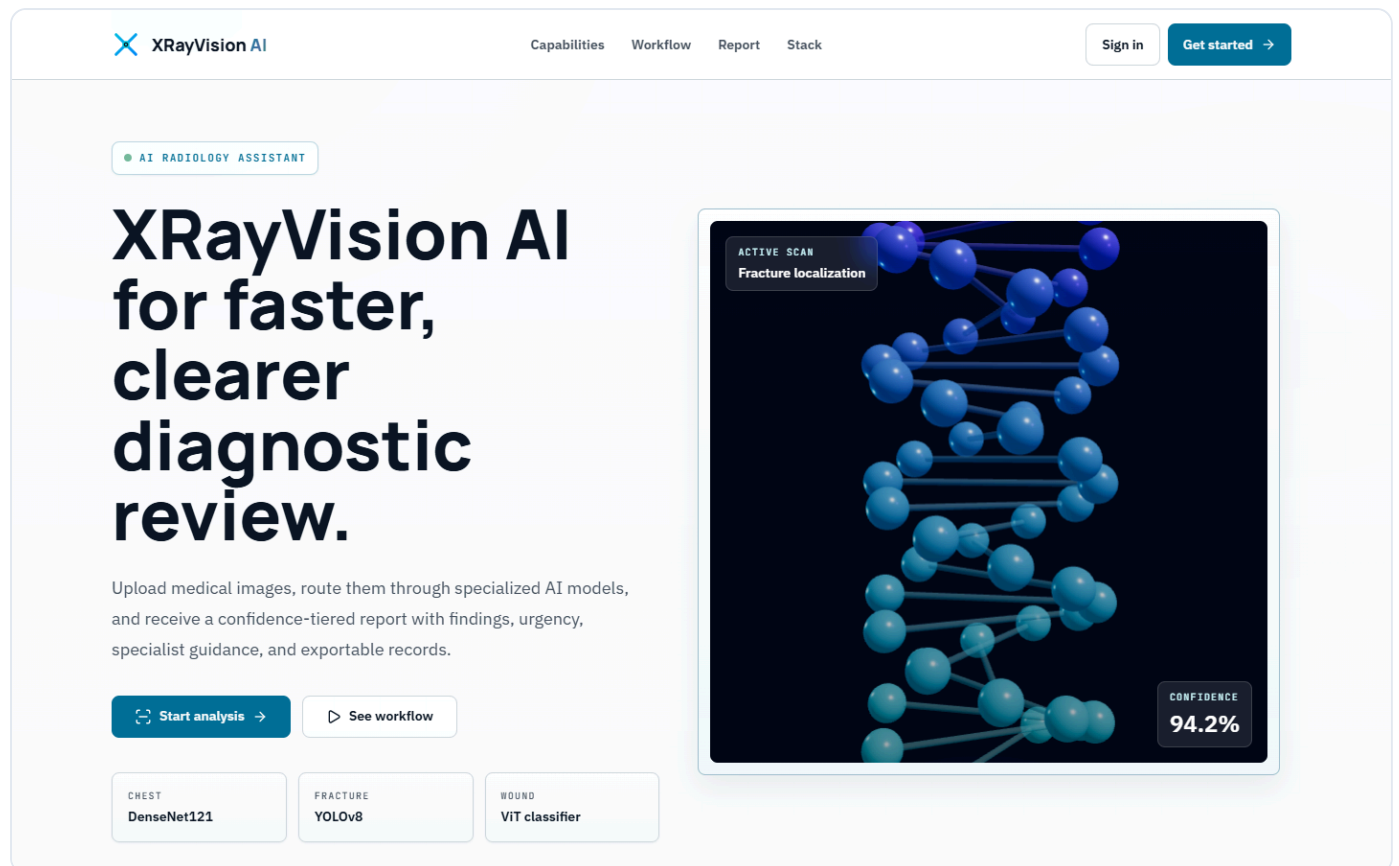


Figure 7.1 — Public landing page.

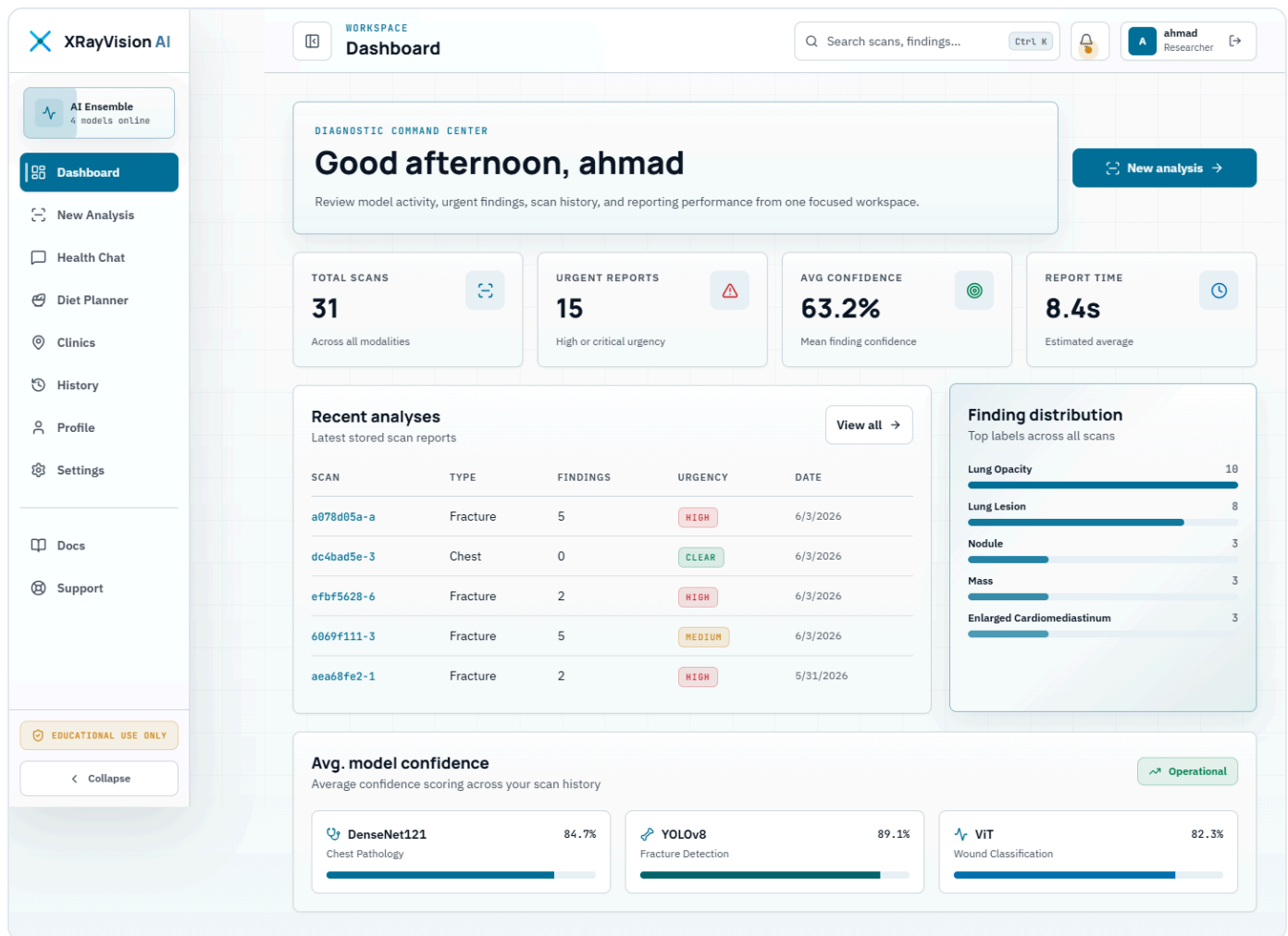


Figure 7.2 — Dashboard showing per-user analytics, recent analyses, finding distribution and per-model confidence.

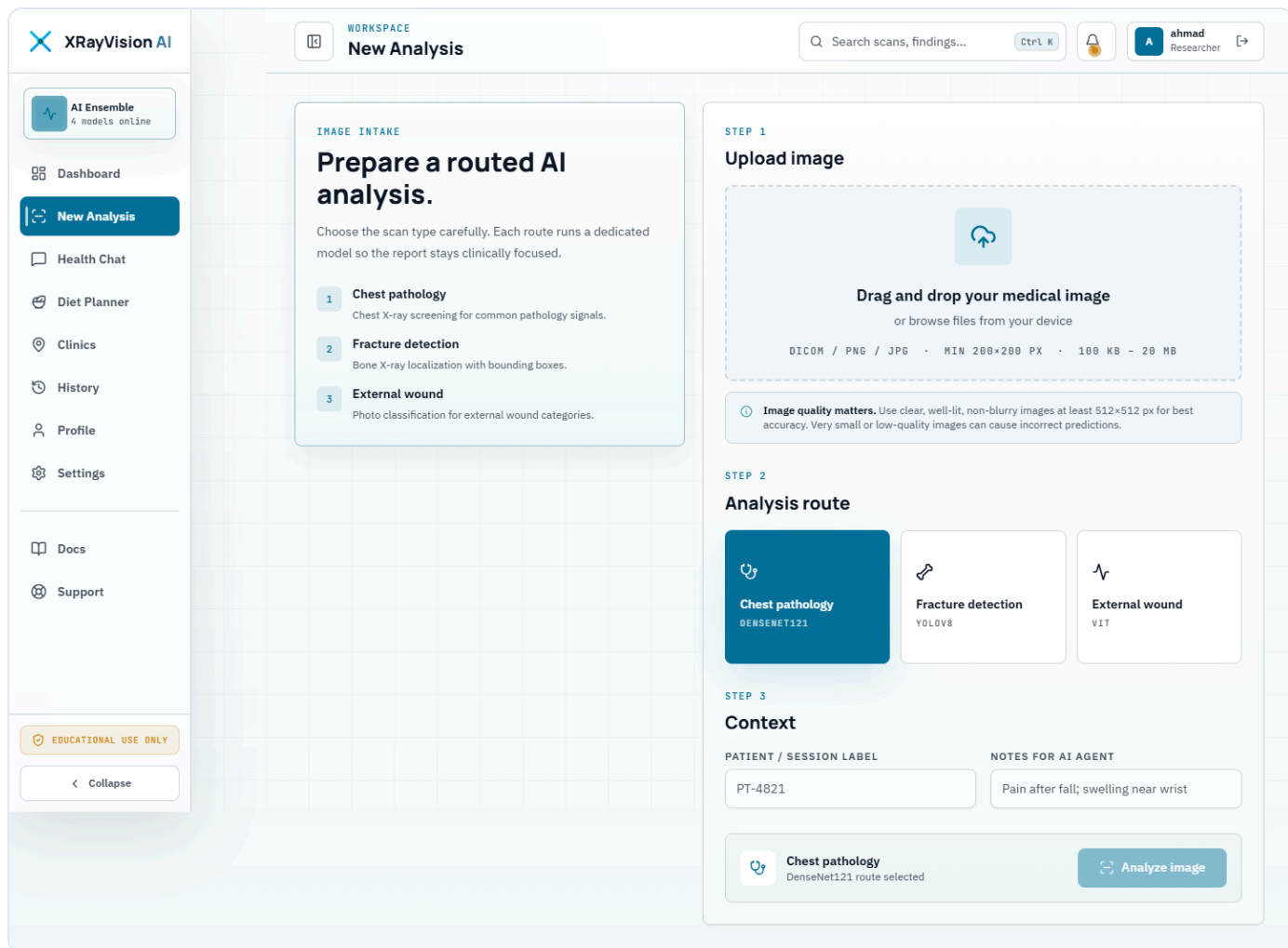


Figure 7.3 — Three-step analysis wizard with format and size constraints stated on screen.

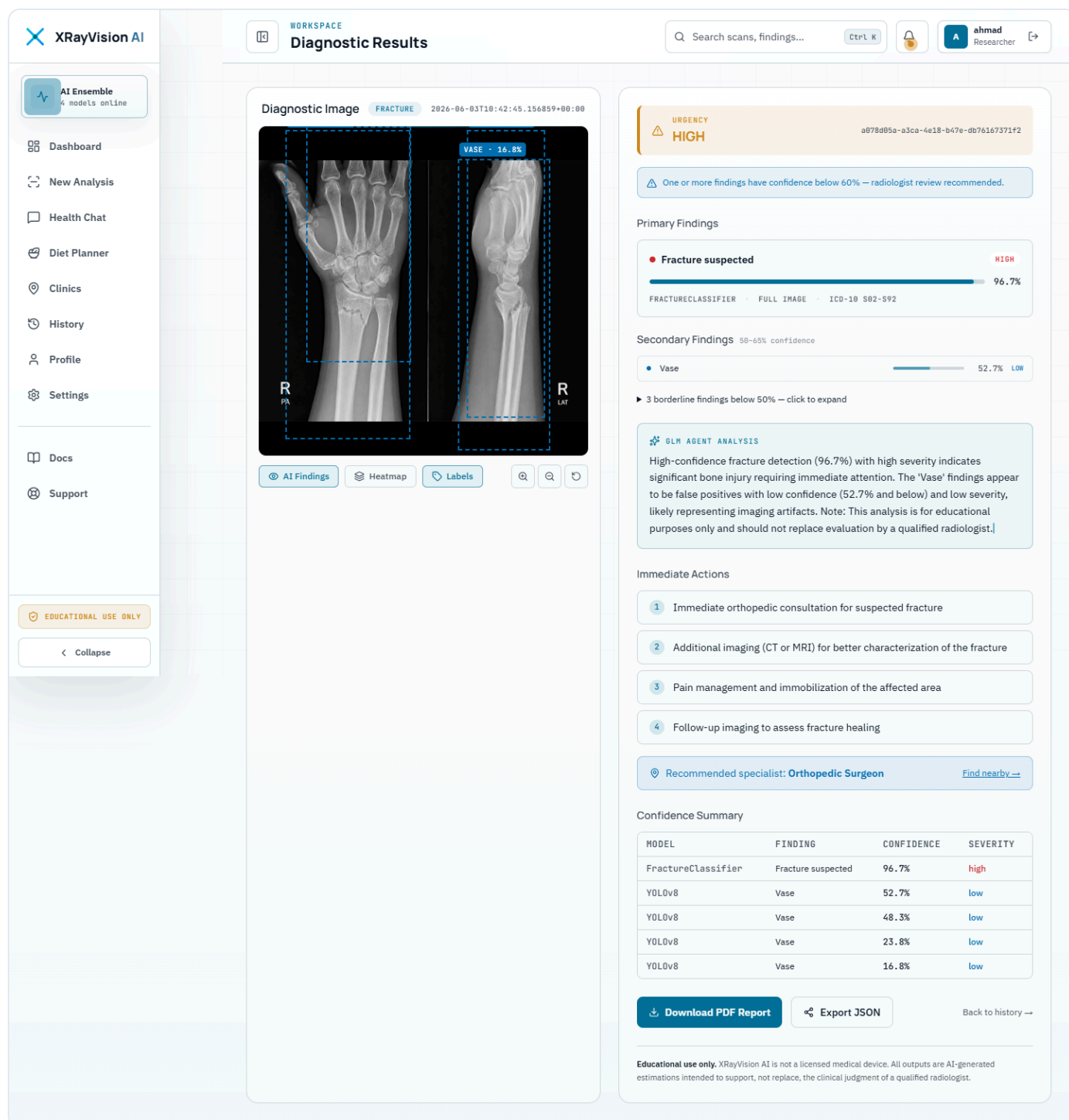


Figure 7.4 — Diagnostic report: bounding-box overlays, tiered findings, low-confidence review banner, LLM synthesis, recommended actions, specialist referral and full confidence table.

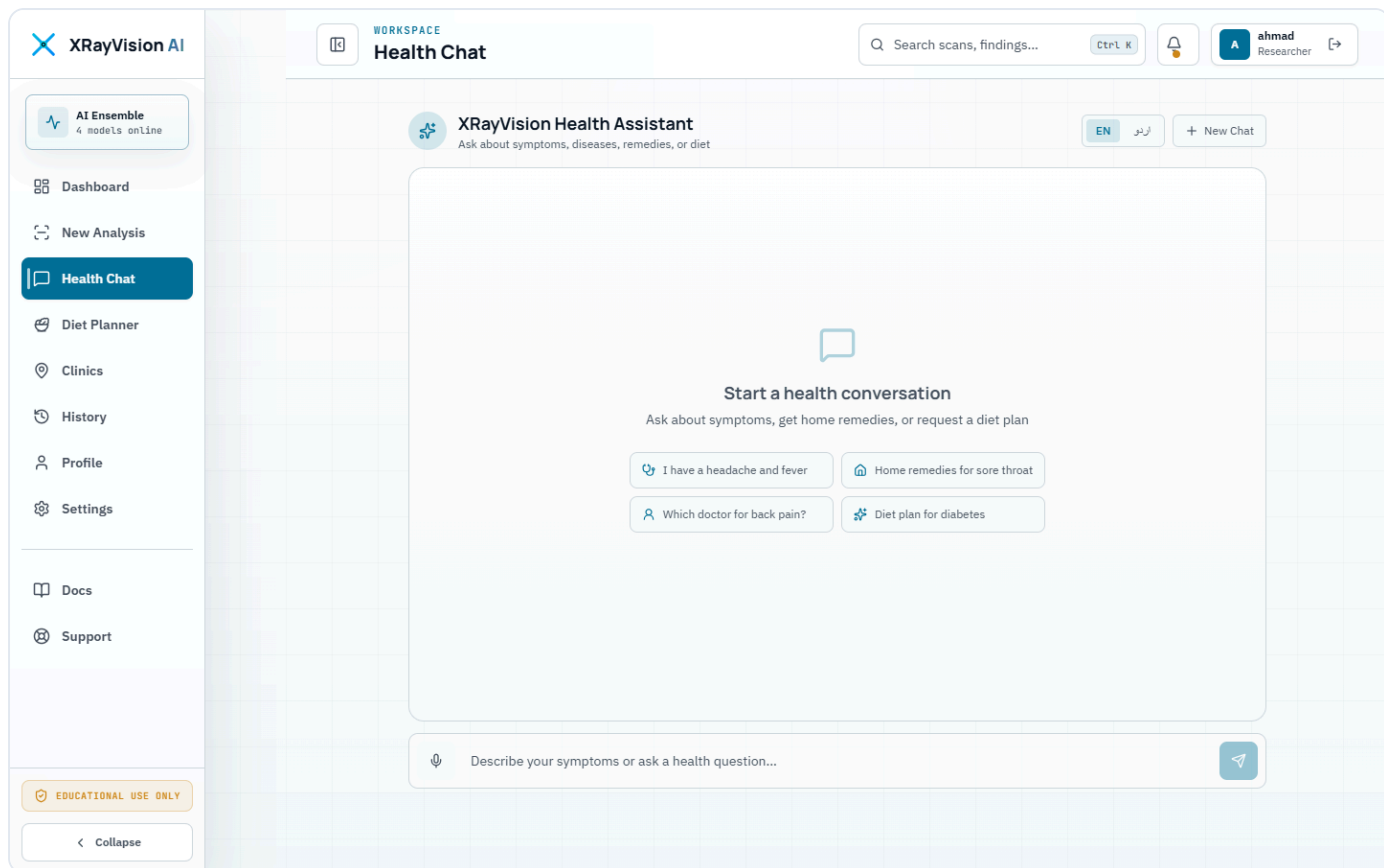


Figure 7.5 — Bilingual health chatbot with English/Urdu toggle.

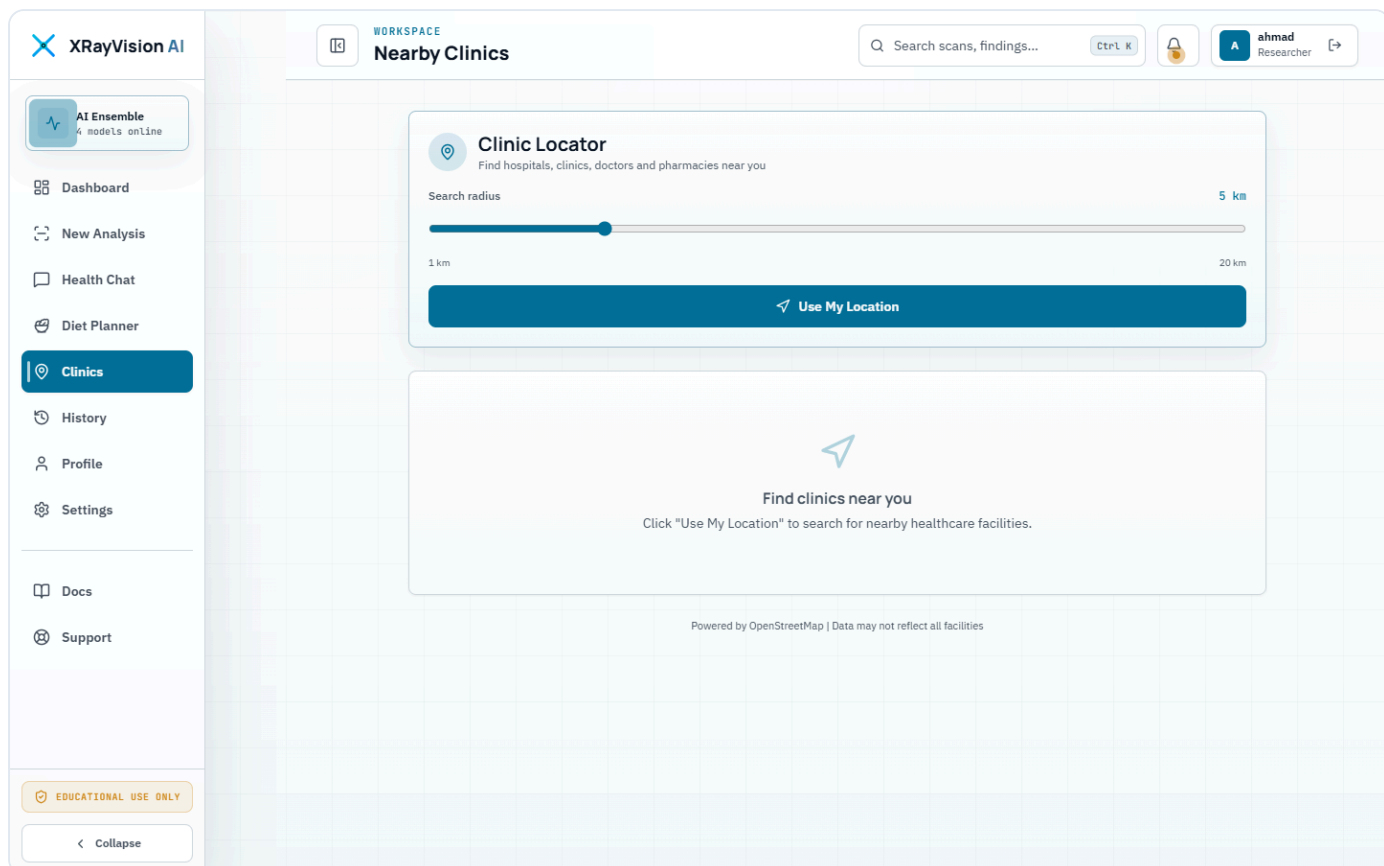


Figure 7.6 — GPS clinic locator with radius control and OpenStreetMap attribution.

Figure 7.4 is worth reading closely, because it demonstrates several design decisions operating together. The scan carries a **HIGH** urgency badge with a text label, not colour alone (D-10, NFR-Q11). A blue banner warns that findings below 60 % confidence are present. The Primary tier contains a single high-confidence fracture finding at 96.7 % with an ICD-10 code; the Secondary tier contains a 52.7 % finding; three further findings below 50 % are collapsed. The LLM synthesis explicitly identifies the low-confidence findings as probable false positives and closes by stating that the analysis is educational and should not replace evaluation by a qualified radiologist. A full confidence table lists every finding with its originating model. It also shows DEF-03: the spurious `vase` labels are visible in the Secondary and Borderline tiers. We have not cropped them out of this thesis.

7.2 Achievement Against Objectives

Table 7.2 — Objectives achieved against Chapter 1 criteria

#	Objective	Criterion	Outcome
O1	Routed ensemble	Correct dispatch; wrong model never run	Achieved. Three routes dispatch correctly; verified in TC-ANLZ-02
O2	LLM synthesis	Urgency, prose, actions, specialist on every analysis	Achieved. Present on all stored scans; fallback path preserves the analysis on failure
O3	Visible uncertainty	Tiering plus sub-60 % review flag	Achieved. Verified in TC-SCAN-04 and TC-SCAN-05
O4	Supporting features	History, exports, analytics, chatbot, diet, clinics	Achieved. All implemented and deployed; export payloads pending verification
O5	Data isolation	No cross-user access through any endpoint	Partially verified. Implemented at two layers; cross-account test (TC-SEC-04) pending execution
O6	Zero-cost deployment	Free tiers, CPU-only	Achieved. No expenditure incurred
O7	Professional documentation	SRS, SDD, tests, manual; defects reported	Achieved. All four documents delivered; three defects reported openly

Six of seven objectives were fully achieved. O5 is marked *partially verified* rather than achieved: the two-layer isolation mechanism is implemented and the code path is straightforward, but the adversarial cross-account test has not been executed, and we do not claim verification we have not performed.

7.3 Discussion

On routing versus ensembling. The decision to route rather than ensemble (D-1) was made for performance under a CPU-only constraint, but its more valuable consequence turned out to be diagnostic focus. Running a chest pathology model on a wound photograph produces output that is not merely

useless but actively misleading — plausible-looking labels with real confidence scores. Routing eliminates a whole class of confusing output.

On the LLM as a translation layer. The synthesis layer is the component that turns model output into something a student can act on. Constraining it to reason only over findings it is given — never over the image — keeps it a translator rather than a second, unvalidated diagnostician. The fallback path (D-3) proved its worth in practice: free-tier LLM availability is intermittent, and without the fallback a substantial fraction of analyses would have failed outright.

On making uncertainty structural. The confidence tiering was designed to communicate uncertainty to users. It turned out to also contain model errors. When DEF-03 caused spurious labels to enter the findings list, tiering automatically relegated them below the correct finding. A flat sorted list would have presented them with equal visual weight. Design decisions made for user comprehension had a defensive benefit we did not anticipate.

On the cost of free infrastructure. Zero cost is not free of consequence. Cold starts add 30–60 seconds to the first request after inactivity. Memory limits forced the `DISABLE_PRELOAD` path. LLM quota required rate limiting. Every one of these constraints shaped the architecture, and the resulting design is arguably more robust for having been forced to assume that dependencies fail.

On what we did not do. We did not fine-tune the models, and we did not evaluate their diagnostic accuracy. Both were deliberate. Fine-tuning without a properly labelled, clinically supervised dataset would risk degrading models trained on vastly more data than we could assemble, while creating a false impression of domain adaptation. Claiming accuracy figures without a validation set and clinical supervision would be worse — it would be the exact overclaiming that makes medical AI tools hazardous.

7.4 Comparison with Related Work

Relative to the systems surveyed in §2.7, this work is distinguished by breadth of integration rather than depth of any single component. CheXNet-class research systems exceed our chest component in rigour of evaluation, but are not deployable applications. TorchXRaiVision provides the model we use but no application layer. Commercial systems offer clinical validation we cannot match, at a cost that excludes the users we target. General-purpose LLM chat interfaces accept images but bring no specialist clinical model and present no calibrated confidence.

Our contribution is the working combination: three imaging modalities, correct routing, LLM synthesis, structurally visible uncertainty, and public deployment at zero cost.

7.5 Ethical and Safety Considerations

Framing. The system is presented as an educational tool throughout — in the application shell, on every report, in the repository README and in this thesis. This framing is not a disclaimer bolted on at the end; it determined design decisions D-5, D-10 and D-11.

The false-negative hazard. The most dangerous outcome is a user concluding from a `clear` result that a study is normal. Three mitigations address this: negative detector boxes are discarded rather than reported as normality (D-5); confidence is always displayed; and low-confidence results trigger an explicit review recommendation. None of these eliminates the hazard. It is inherent to the category of tool, and honesty about it is the only responsible position.

Privacy. No patient-identifying information is requested or stored. Storage paths are keyed by opaque UUIDs. Users may delete any scan permanently, and account deletion cascades to all associated data. We note against this that the storage buckets are public (D-15) — an unguessable URL is not the same as an access-controlled one, and §8.3 lists this as a limitation to be remediated.

Automation bias. A fluent, confident paragraph is persuasive independently of correctness. The system counters this by always displaying the numerical confidences the prose was derived from, by tiering findings, and by instructing the language model to state its own limitations — which, as Figure 7.4 shows, it does.

Language access. Urdu support with right-to-left rendering and voice input was included because health information tools in Pakistan are overwhelmingly English-only. The same safety constraints apply to both languages.

Chapter 8: Conclusion and Future Work

8.1 Conclusion

This thesis presented XRayVision AI, a deployed web platform that integrates three clinically pretrained vision models and a large language model into an educational medical imaging tool operating entirely on free-tier infrastructure with CPU-only inference.

The work set out to close an accessibility gap. Capable medical imaging models are freely available, but they are research artefacts rather than usable systems, they present output in a form learners cannot act on, they rarely communicate uncertainty adequately, and the infrastructure normally assumed around them is beyond the reach of a teaching institution. The delivered system addresses each of these: it provides the full application layer, translates model output into clinical prose through a constrained LLM, makes uncertainty structurally visible through three confidence tiers and an explicit review flag, and does so at zero cost.

Two design principles proved more valuable than anticipated. **Safety by construction** — encoding safety into structure rather than into warnings — meant that when a model produced incorrect output (DEF-03), the confidence tiering automatically contained it. **Graceful degradation** — treating dependency failure as a reason to downgrade rather than to abort — meant the system remained useful on infrastructure that fails routinely.

The system was specified, designed and verified to professional standards, and its limitations are documented rather than concealed. Seventeen of ninety-three test cases were executed and three defects remain open, including one substantive semantic defect in the fracture detector's label handling. We report these because a thesis that presents only what worked teaches less than one that also reports what did not.

8.2 Contributions

1. **An integrated, publicly deployed multi-modal medical imaging platform** combining chest pathology screening, fracture localisation and wound classification with LLM report synthesis.
2. **A routed ensemble architecture** that dispatches by declared scan type rather than executing all models, making multi-model inference viable on CPU-only free-tier infrastructure.
3. **A confidence tiering presentation** that makes model uncertainty structurally visible and, as demonstrated in §7.3, provides incidental containment of model error.

- 4. **An explicit negative-evidence safety rule** — that a detector's absence of detection is never reported as normality — implemented and verified.
- 5. **A degradation architecture** in which the failure of the language model, of object storage or of any individual vision model downgrades rather than destroys the result.
- 6. **A complete professional documentation set** (IEEE 830, 1016 and 829) with honest reporting of verification status and open defects.
- 7. **Bilingual English/Urdu health information access** with right-to-left rendering and voice input.

8.3 Limitations

- 1. Diagnostic accuracy was not independently validated; no clinical performance claim is made.
- 2. DEF-03 — non-anatomical detector labels reach the findings list — is unresolved.
- 3. Seventy-five of ninety-three test cases remain unexecuted.
- 4. Storage buckets are public; URL obscurity is not access control.
- 5. Rate limiting is per-IP and in-memory, so it resets on restart and is shared by users behind NAT.
- 6. No refresh-token rotation; users must re-authenticate every 24 hours.
- 7. No audit log; deletions are not traceable.
- 8. Uploaded files are not scanned for malicious content before decoding.
- 9. Cold starts add 30–60 seconds to the first request after inactivity.
- 10. Cross-browser verification was performed on Chromium only.

8.4 Future Work

Table 8.1 — Future work priorities

Priority	Item	Rationale
Immediate	Resolve DEF-03: correct the deployed <code>YOLO_WEIGHTS_PATH</code> , and validate the loaded model's class vocabulary at startup rather than trusting its filename	Incorrect labels currently reach users
Immediate	Execute the remaining 75 test cases	Verification is incomplete
Immediate	Close DEF-01 and DEF-02	Repository hygiene and documentation correctness
High	Migrate storage to private buckets with short-lived signed URLs	Removes reliance on URL obscurity
High	Add an automated test suite (pytest, Vitest) integrated into CI	Currently no automated regression protection exists
High	Clinical validation against a labelled evaluation set under supervision	Required before any accuracy claim
Medium	Redis-backed rate limiting keyed by user identity	Survives restarts; fairer behind NAT

Priority	Item	Rationale
Medium	Refresh-token rotation	Removes the 24-hour re-authentication burden
Medium	Append-only audit log	Traceability for deletions
Medium	Explainability overlays (Grad-CAM) for the chest model	Localisation for chest findings, matching the fracture route
Medium	Malware scanning of uploads before decoding	Defence against crafted files
Lower	GPU inference tier for reduced latency	Requires funding
Lower	Additional imaging modalities (CT, MRI)	Substantial scope extension
Lower	PACS / DICOM-network integration	Required for any real clinical deployment
Lower	Additional regional languages	Broadens access further

8.5 Closing Remarks

The most useful lesson from this project was not technical. It was that in a domain where being wrong has consequences, the discipline of designing for uncertainty and for failure produces a better system than designing for the success case and adding warnings afterwards. The confidence tiering that contained a model defect, and the fallback path that kept the system useful through intermittent LLM availability, were both consequences of assuming things would go wrong.

We hope the system is useful to students learning to read radiographs, and that the documentation — including its account of what did not work — is useful to anyone attempting similar work.

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Appendix A: System Screenshots

Additional interface screenshots captured from the live deployment.

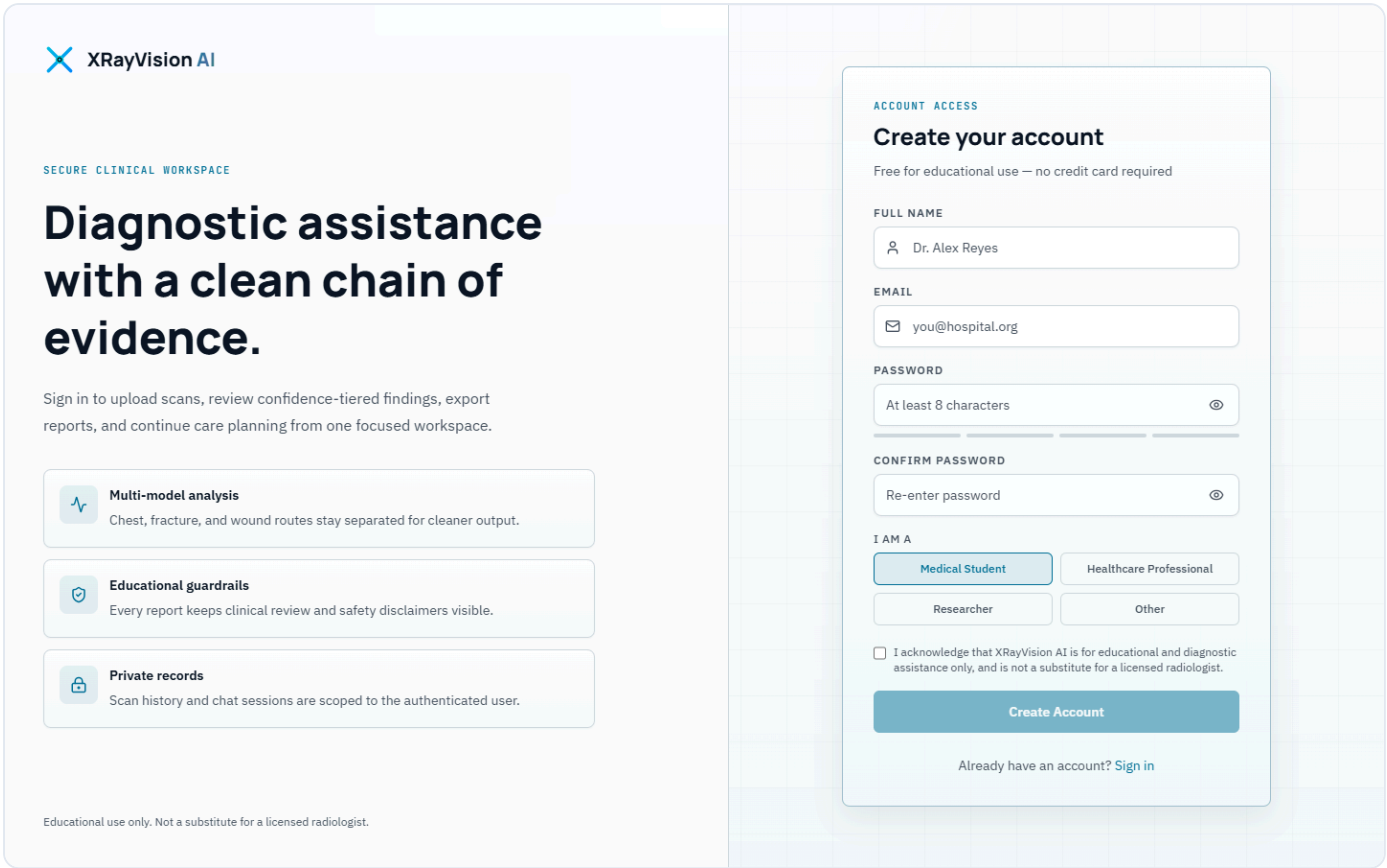


Figure A.1 — Account registration.

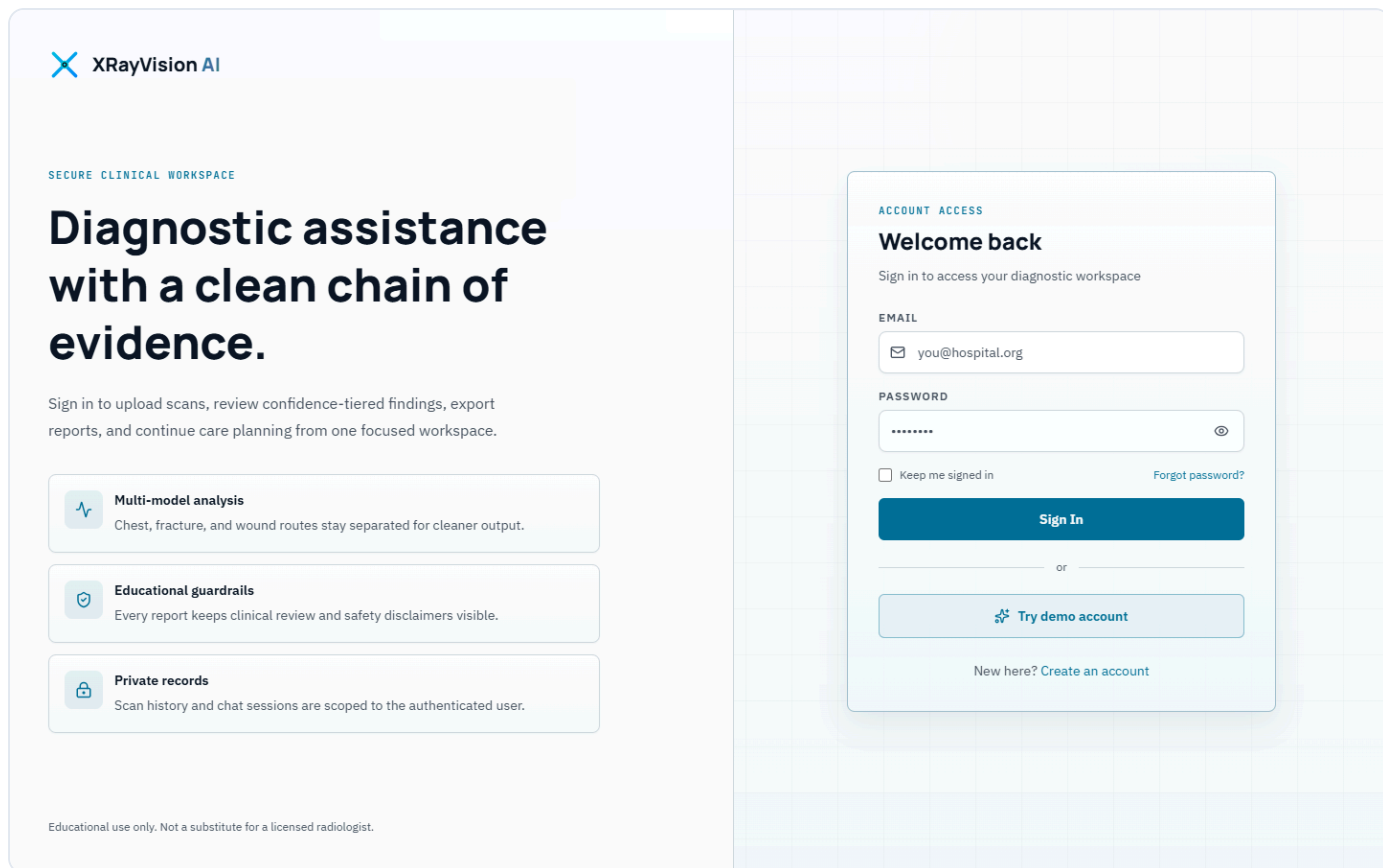


Figure A.2 — Sign-in, including the one-click demonstration account.

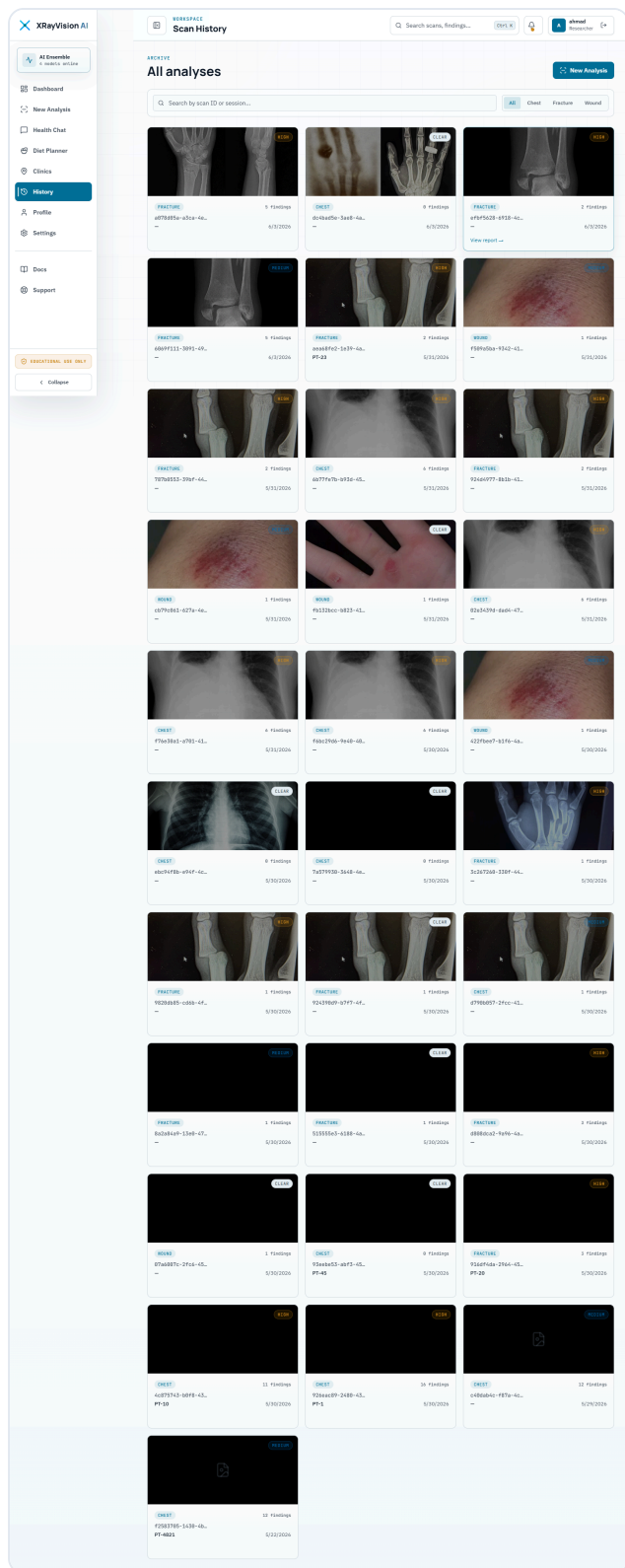


Figure A.3 — Scan history.

WORKSPACE

Diet Planner

Ctrl K

ahmad
Researcher

AI Ensemble
4 models online

Dashboard

New Analysis

Health Chat

Diet Planner

Clinics

History

Profile

Settings

Docs

Support

EDUCATIONAL USE ONLY

< Collapse

AI Diet Plan Generator

Get a personalized meal plan based on your health condition

MEDICAL CONDITION (OPTIONAL)

DIETARY PREFERENCE

Balanced

Vegetarian

Vegan

Keto

High-Protein

Low-Carb

RESTRICTIONS (COMMA-SEPARATED)

HEALTH GOALS

English Urdu

⚠ Educational meal guidance only. For hypertension, diabetes, kidney disease, pregnancy, allergies, or prescribed medicines, review the plan with a qualified clinician or dietitian.

Generate Diet Plan

Figure A.4 — Diet plan generator.

WORKSPACE

Profile

Ctrl K

ahmad
Researcher

AI Ensemble
4 models online

Dashboard

New Analysis

Health Chat

Diet Planner

Clinics

History

Profile

Settings

Docs

Support

EDUCATIONAL USE ONLY

< Collapse

ACCOUNT PROFILE

ahmad

ahmad123@gmail.com

Member since 5/22/2026

ROLE

Researcher

SCANS

31

AVG CONFIDENCE

63.2%

Personal details

Keep your identity visible in reports and workspace activity.

FULL NAME

ahmad

EMAIL

ahmad123@gmail.com

ROLE

Researcher

Save Changes

Usage Statistics

TOTAL SCANS

31

CRITICAL FINDINGS

15

AVG. CONFIDENCE

63.2%

Session

Sign out of your account on this device.

Sign out

Figure A.5 — User profile.

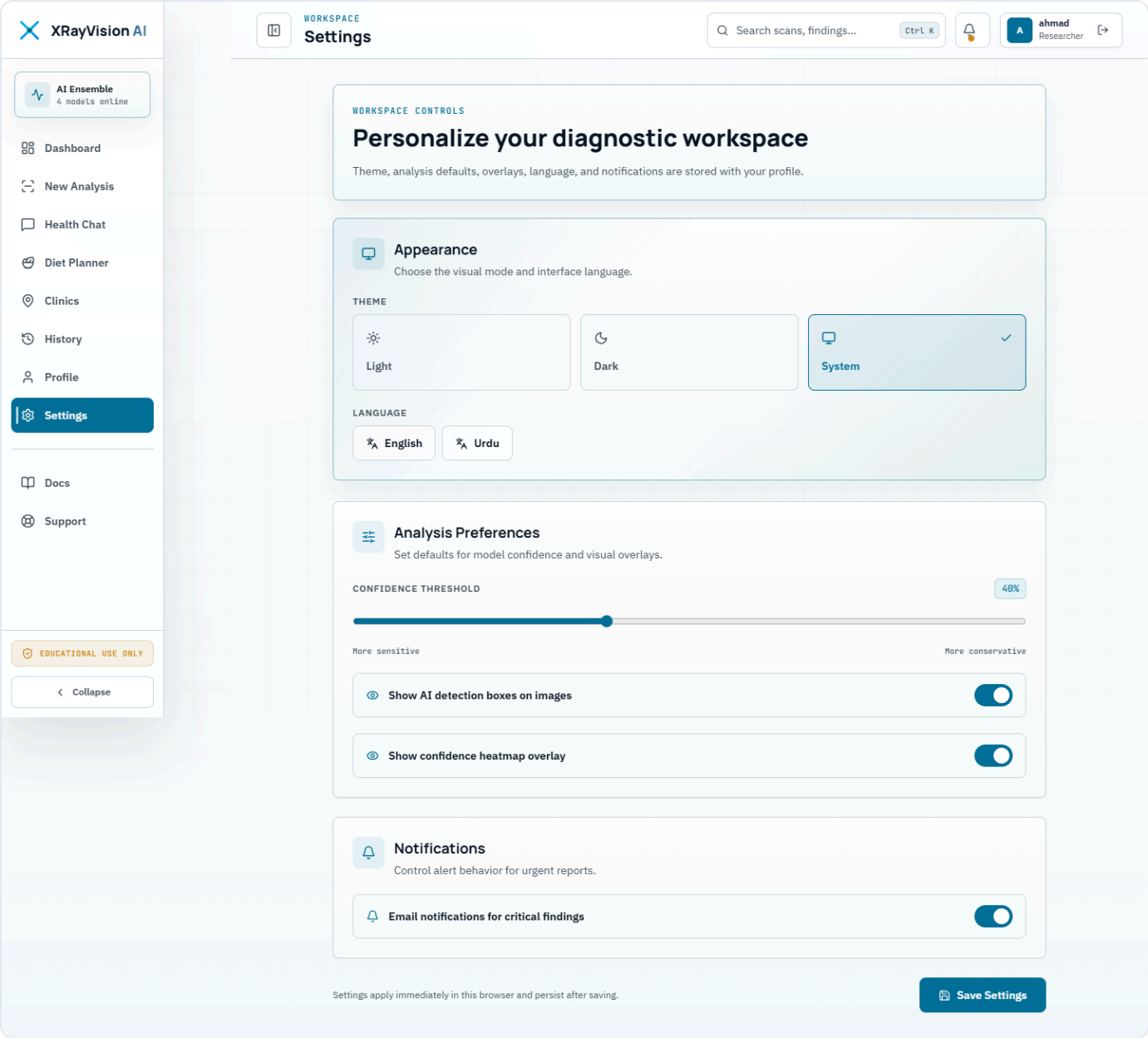


Figure A.6 — Settings: appearance, analysis preferences and notifications.

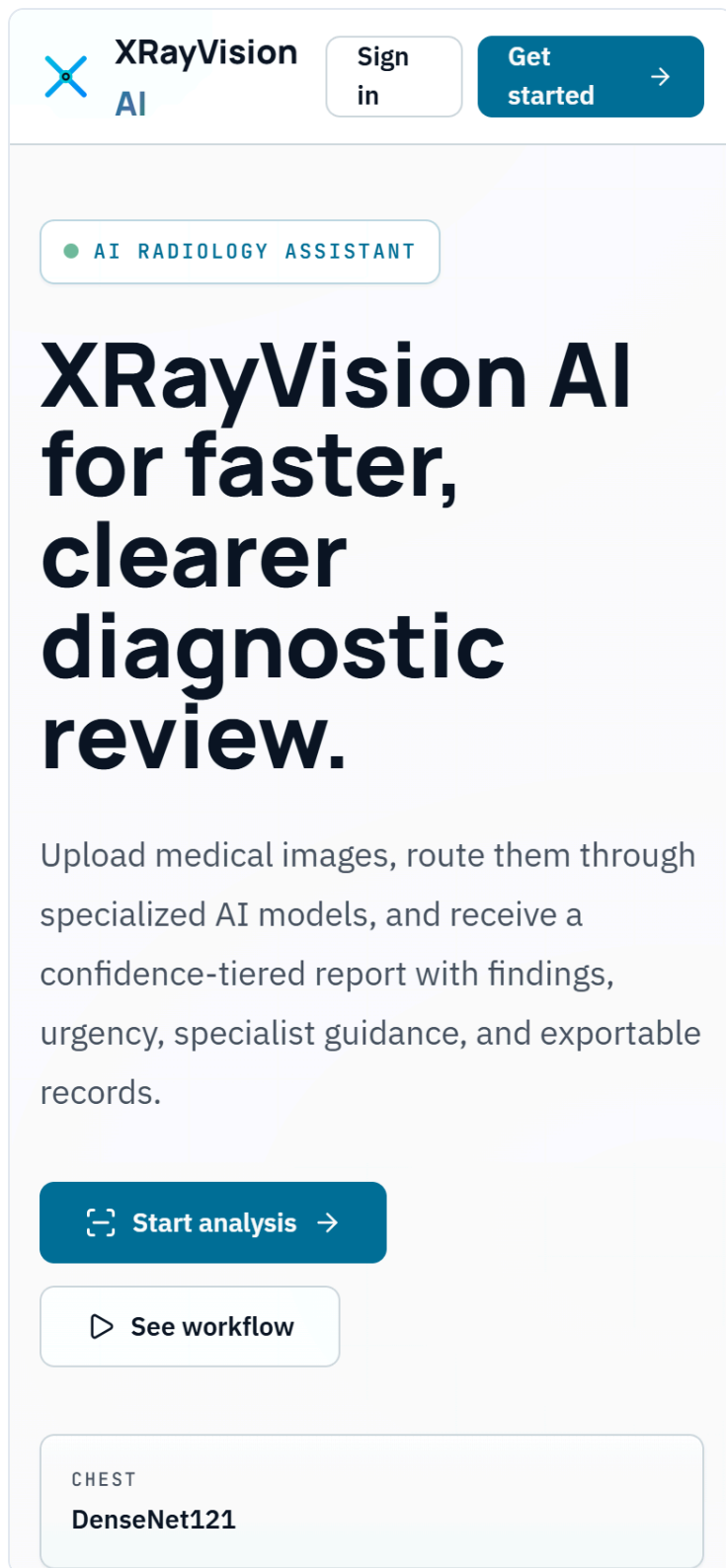


Figure A.7 — Responsive layout at 390 × 844 pixels.

Appendix B: API Reference

Table 5.3 — REST API endpoint summary

#	Method	Endpoint	Auth	Rate limit	Purpose
1	POST	/auth/register	—	200/min	Create an account
2	POST	/auth/login	—	200/min	Authenticate, return JWT
3	POST	/auth/forgot-password	—	200/min	Trigger a reset email
4	GET	/auth/me	JWT	200/min	Current user profile
5	POST	/auth/avatar	JWT	200/min	Upload a profile picture
6	PATCH	/auth/profile	JWT	200/min	Update name and role
7	PATCH	/auth/settings	JWT	200/min	Update preferences
8	POST	/analyze	JWT	10/min	Upload and analyse an image
9	GET	/scans	JWT	200/min	List scan history
10	GET	/scans/{id}	JWT	200/min	Retrieve one scan
11	DELETE	/scans/{id}	JWT	200/min	Delete a scan and its image
12	GET	/scans/{id}/report.pdf	JWT	200/min	Download a PDF report
13	GET	/scans/{id}/export.json	JWT	200/min	Export raw findings
14	POST	/chat	JWT	20/min	Send a chatbot message
15	GET	/chat/sessions	JWT	200/min	List chat sessions
16	GET	/chat/sessions/{id}/messages	JWT	200/min	Retrieve a conversation
17	POST	/diet	JWT	10/min	Generate a diet plan
18	GET	/stats	JWT	200/min	Dashboard statistics
19	GET	/clinics	JWT	200/min	Find nearby facilities
20	GET	/	—	200/min	Service metadata
21	GET	/health	—	200/min	Health check

Interactive documentation is generated automatically and available at `/docs` (Swagger UI) and `/redoc` (ReDoc) on the running backend.

Appendix C: Database Schema

The complete schema is provided in `backend/supabase_schema.sql` . Its structure is summarised below.

Tables. `profiles` (extends `auth.users` one-to-one), `scans` , `chat_sessions` , `chat_messages` .

Constraints. `scans.scan_type` is constrained to `chest | fracture | wound` ; `scans.urgency` to `critical | high | medium | low | clear` ; `chat_messages.role` to `user | assistant` . Every foreign key referencing `profiles` specifies `ON DELETE CASCADE` .

Indexes. `idx_scans_user_id` , `idx_scans_created_at` (descending), `idx_chat_sessions_user_id` , `idx_chat_messages_session_id` .

Row Level Security. Enabled on all four tables. `SELECT` policies restrict access to rows owned by `auth.uid()` ; `INSERT` is performed by the backend service role after independent JWT verification.

Trigger. `handle_new_user()` fires `AFTER INSERT ON auth.users` and creates the corresponding `profiles` row, defaulting `role` to `Medical Student` .

Storage buckets. `xray-images` (path `{user_id}/{scan_id}.{ext}`) and `avatars` (path `{user_id}.{ext}`), both created idempotently at first use.

Appendix D: Configuration Reference

Variable	Default	Purpose
SUPABASE_URL	—	Supabase project URL
SUPABASE_KEY	—	Service-role key (server-side only)
SUPABASE_ANON_KEY	—	Public anon key
OPENROUTER_API_KEY	—	LLM API key
OPENROUTER_MODEL	z-ai/glm-4.5-air:free	LLM model identifier
OPENROUTER_TIMEOUT_SECONDS	60	LLM request timeout
HF_TOKEN	—	Hugging Face read token
JWT_SECRET	—	HS256 signing secret, minimum 32 characters
JWT_ALGORITHM	HS256	Token signing algorithm
JWT_EXPIRY_HOURS	24	Token lifetime
YOLO_WEIGHTS_PATH	models/fracture_yolov8.pt	Fracture detector weights
ALLOW_GENERIC_YOLO_WEIGHTS	false	Guard against non-medical checkpoints — see DEF-03
FRACTURE_CLASSIFIER_ENABLED	true	Enable image-level fracture screening
FRACTURE_CLASSIFIER_MODEL_NAME	prithivMLmods/Bone-Fracture-Detection	HF fracture classifier
WOUND_MODEL_NAME	PayamFard123/dermaintel-wound-classifier	HF ViT wound classifier
CONFIDENCE_THRESHOLD	0.40	Minimum confidence for a reported finding
FRONTEND_URL	http://localhost:5173	Primary CORS origin
ALLOWED_ORIGINS	—	Additional comma-separated production origins
DISABLE_PRELOAD	false	Skip preloading on memory-constrained hosts

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Educational use only.

This system is not a certified medical device.

All outputs are AI-generated estimates intended to support, not replace, the clinical judgement of a qualified radiologist.